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On the applicability of time series anomaly detection methods to real-world traffic volume data

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ABSTRACT

Time Series Anomaly Detection (TSAD or TAD) refers to the automatic and data-driven identification of abnormal segments in time series data, a task that has been studied extensively for decades. Despite recent transformative and novel findings revealed by efforts in this field, the literature on traffic anomaly detection has not yet fully reflected on these emerging trends to draw practical conclusions. In this paper, we focus on the applicability of state-of-the-art and well-established TAD methods to road traffic volume data, making contributions in two main ways. First, given the proven and major contribution of evaluation data to TAD outcomes, we argue that existing anomaly-labeled datasets from transportation and traffic systems require substantial enhancements in terms of both data size and label quality. To address this, we propose a new platform to inspect and label large-scale volume data of urban areas based on its unique characteristics and the latest taxonomy of time series anomalies. Second, based on the established framework, we also formulate the TAD problem in traffic volume data and introduce a discord-based, context-embedded, and light-weight traffic anomaly detection method, named Step-isolated Traffic Discords Discovery (Si-TDD), to address this problem. Benefiting from our labeling platform, AnoLT (Anomaly Labeled Traffic) is presented in this paper for the first time as a comprehensive, open-source, and anomaly-labeled spatiotemporal dataset collected from 147 locations across Melbourne, Australia. Comparative results with more than 20 baselines also indicate that Si-TDD considerably outperforms recent TAD solutions when it comes to traffic volume data, achieving a 67% F1 score with the AnoLT dataset. This paper highlights the key role of incorporating context-related information into existing TAD solutions to boost their effectiveness in traffic anomaly detection, a factor that is often overlooked in the current literature.

1. Introduction

Road traffic count/volume time series data is among the most vital resources used in the management of traffic networks. Automatic detection of outliers (or anomalies) in this data is essential, as it can contribute to both the safety and efficiency of the system (Wang and Sun, 2021; Sarteshnizi et al., 2023). Traffic anomaly detection not only provides valuable insights for understanding and managing traffic performance in both real-time and offline situations but also helps maintain the reliability of information recorded from detectors, which is essential for effective traffic modeling and prediction.

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Different from incident detection studies, which mostly identify operational disruptions in networks continuously equipped with detectors and mainly rely on traffic speed information (Kalair and Connaughton, 2021; Chakraborty et al., 2019), a growing body of work is emerging that examines traffic data anomalies from a more "data-centric" perspective. This line of research, including the present study, aims to detect all data inconsistencies, defined as segments that considerably deviate from the dominant historical patterns of the data, regardless of their underlying causes or whether they genuinely correspond to traffic incidents (Sarteshnizi et al., 2023; Deng et al., 2022). Although detecting operational events in traffic data (such as incidents) is valuable, it should be noted that traffic data can exhibit a variety of anomaly types, including sharp point spikes or drops, sudden or moderate subsequence changes, and/or prolonged deviations from typical profiles, which can, in turn, provide additional insights or even complicate event detection processes. Such inconsistencies can be caused by a variety of factors, including, but not limited to, incidents, extreme weather, faulty sensors, and unpredictable demand fluctuations. Automatic detection of data-driven anomalies also creates opportunities for a wide range of downstream tasks. These range from revealing hidden patterns in big data and supporting analysts with easy access to suspicious intervals during exploratory analyzes to managing data quality for automated modeling or simulation pipelines and enabling operational triggers when certain types of patterns are observed. Therefore, detecting such anomalies in large-scale data in an unsupervised manner, before exploring their causes and consequences, is essential and has recently gained renewed attention, particularly with the rapid growth of research in Time Series Anomaly Detection (Zamanzadeh Darban et al., 2024).

Time Series Anomaly Detection (TSAD or TAD), as a general topic, has been widely investigated for many years, resulting in various solutions to tackle its unique challenges (Schmidl et al., 2022). Recently, the progress in this area has uncovered some fundamentally flawed assumptions used in the evaluation processes of the state-of-the-art works (Nakamura et al., 2023; Kim et al., 2022; Hwang et al., 2022). These discoveries are gaining prominence, emphasizing the importance of adopting new perspectives in "evaluation pipelines" rather than solely focusing on "methodological" advancements (Lai et al., 2021b). The main component criticized in these pipelines is the *data* itself, as it has been shown that, with new setups of data, no superiority can be claimed for advanced methods over classical algorithms (Liu and Paparrizos, 2024). In both supervised and unsupervised setups, practitioners require labeled data to validate hypothetical solutions. However, obtaining such data is challenging, particularly because the exact occurrences of anomalies are often unknown. Although several real-world TAD benchmarks exist, some of which are related to urban traffic, they have been found to deviate significantly from the ideal standard. For instance, Wu and Keogh (2021) showed that many of these datasets, including the NYC taxi demand (Ahmad et al., 2017), are flawed and create an illusion of progress. As a result, researchers have begun developing new datasets, often synthesized, to address these evaluation shortcomings (Wang and Sun, 2021). This focus on synthesized datasets is also resulting in a limited understanding of how proposed solutions perform in real-world scenarios. Anomalies in real-world data are inherently more complex due to the presence of noise, various data qualities, and diverse levels of variability. According to recent benchmarks, performance indicators typically experience a significant drop when existing solutions are evaluated with real-world datasets (Paparrizos et al., 2022; Lai et al., 2021b).

In addition to these data-related challenges, which remain prevalent and require attention, other points arise from studies specifically focused on traffic data anomaly detection. First, the evaluation data in a number of works comes from private resources and cannot be leveraged in future efforts (Kalair and Connaughton, 2021; Parsa et al., 2019). Second, they often use incident reports or public event information as ground truth labels for traffic data. This introduces new challenges in evaluation experiments, as public resources do not report on all existing anomalies (Chakraborty et al., 2019; Sarteshnizi et al., 2023). Third, baseline methods employed in urban traffic domain are usually incompatible across different works, resulting in varied claims and a lack of clear conclusions regarding the best method (He et al., 2023; Wang and Sun, 2021; Deng et al., 2022). Researchers often focus on specific perspectives for anomaly detection, with a need for greater elaboration on the rationale behind these approaches and their added value compared to previous works. And last but not least, fewer efforts can be found bridging the gap between the findings of general TAD studies and urban traffic perspectives. Although there has been a renewed interest in classical univariate methods in TAD stemming from a revision of the problem and definitions (Nakamura et al., 2023; Liu and Paparrizos, 2024), their transferability has not yet been fully explored in the context of traffic data. On the other side, despite the availability of various urban traffic datasets in TAD (Zamanzadeh Darban et al., 2024), the quality and suitability of these representative datasets is yet to be discussed from traffic engineering perspectives.

To move forward in above research gaps, we focus on exploring the applicability of TAD methods to traffic volume data and elaborate on *univariate* modeling as this has not yet been explored. The key reasons for this are twofold. First, despite being recognized as a competitive and even more successful type of TAD solution (Nakamura et al., 2023; Tafazoli and Keogh, 2023), there is limited understanding of the performance of univariate solutions when applied to large-scale and real-world traffic data (Sarteshnizi et al., 2023). Substantial evidence suggests that generalizing univariate approaches to multivariate settings offers a more robust pathway for addressing TAD across various contexts, as it enables more insightful, scenario-inclusive, and lightweight solutions (Wu and Keogh, 2021; Yeh et al., 2024). Second, relevant studies in urban traffic domain often regard deep learning or advanced spatiotemporal methods as the default solutions, neglecting to benchmark and justify their ideas against more customized univariate methods (Deng et al., 2022; Wang and Sun, 2021). Based on the definitions of anomalies so far, anomaly types are always defined solely based on the progression of the series itself, making them theoretically detectable if the univariate pattern is accurately reflected. Therefore, our mission here is to ensure full leverage of existing univariate solutions-especially the well-established ones-is achieved before pursuing complex modeling in traffic time series anomaly detection.

In this work, we aim to establish a base in traffic anomaly detection by first adapting data-driven anomaly definitions and then examining and refining current univariate methods. Building on a recent taxonomy of anomalies in time series data and the characteristics of traffic flow, we first design a data-centric and visual labeling framework to rigorously annotate *all data anomalies* in a large-scale traffic volume dataset. This platform enables the creation of complete ground truth labels for real-world traffic data

in a data-driven way, facilitating their use in comparative studies. Then, we provide a systematic problem formulation to evaluate various state-of-the-art TAD methods in automating the labeling process using the obtained data. Inspired by the time series discord concept, which has shown notable success in many problems (Wu and Keogh, 2021), we also present a customized, lightweight, and effective detection method named Step-isolated Traffic Discords Discovery (**Si-TDD**). This method is then shown and discussed to outperform state-of-the-art baselines from the literature. Benefiting from our labeling platform, we also introduce **AnoLT** (Anomaly Labeled Traffic), a new open-source, real-world, and fine-grained labeled traffic volume dataset.[1] This dataset has been meticulously monitored by two independent experts and offers a new data-driven perspective for TAD evaluation in urban traffic data. AnoLT includes over 3 million labeled data points from 147 road sections located at signalized intersections across Melbourne, Australia. Overall, the key contributions of this paper can be listed as follows:

- To the best of our knowledge, this paper is among the first to discuss the appearance of different time series anomaly types in traffic volume data and to derive complete ground truth labels for them using rigorous observational approaches.
- A new open-source and large-scale traffic volume dataset (AnoLT) with two independent sets of anomaly labels is presented, developed using the latest data-driven taxonomy of anomalies.
- Several anomaly detection methods are evaluated with AnoLT for the first time, discussing the applicability of TAD methods in traffic data and emphasizing the importance of their tailored adaptation prior to deployment in this domain.
- The effectiveness of a customized and lightweight method (Si-TDD) is demonstrated in this work, showing its capability to highlight anomalies in a large, explicitly labeled traffic volume dataset, with performance exceeding that of state-of-the-art machine learning and deep learning methods.

In the rest of the paper, we first discuss recent works in Section 2. Anomaly types, our data labeling framework, and detection methods are explained in Section 3. Results and relevant discussion are provided in Section 4. In the end, a summary of the paper, along with the conclusions, is presented in Section 5.

2. Related works

2.1. Time series anomaly detection (TAD)

Research on anomaly/outlier detection in both univariate and multivariate time series data has been ongoing for several decades (Fox, 1972). It began with benefiting from statistical measures, such as median and thresholds (Basu and Meckesheimer, 2007). Then, more advanced approaches started to emerge regarding specific applications such as smoothing methods (Chen et al., 2010), classical predictive methods (ARIMA) (Akouemo and Povinelli, 2014), and Gaussian Mixture Models (GMM) (Reddy et al., 2017). With the advent of the big data era and the emergence of machine learning and deep learning methods, numerous studies have sought to address TAD using these innovative techniques (Aggarwal, 2016; Du et al., 2017; Liu et al., 2019; Geiger et al., 2020). Despite the advantages attributed to TAD learning-based methods, a new focus of research has arisen emphasizing the evaluation process over the methodology itself. For instance, Wu and Keogh (2021) discussed the quality of existing benchmark datasets, highlighting various issues, including inaccurate label sets and a lack of diversity among them. Kim et al. (2022) also argued that the Point Adjustment method used for alleviating ground truth problems in earlier studies results in an overestimation of performance. Inspired by the results in these papers, researchers have initiated new comprehensive comparative studies to address biases and reassess existing solutions (Lai et al., 2021b; Paparrizos et al., 2022; Nakamura et al., 2023; Liu and Paparrizos, 2024). Findings of these works surprisingly did not demonstrate any meaningful superiority of recent deep learning approaches compared to classical methods in TAD when applied to various applications. In fact, they revealed that TAD outcome varies based on the specific application and data characteristics, with classical methods demonstrating a higher success rate across many different scenarios.

From a *data* perspective, a review of the literature also highlights the critical need for development of new datasets. Yahoo (Laptev et al., 2015), NAB (Ahmad et al., 2017), NASA (Hundman et al., 2018), and OMNI (Su et al., 2019) datasets are among the primary ones in this area. Although several studies benefited from these datasets (Geiger et al., 2020), they suffer from a number of issues such as triviality, mislabeled data, problematic anomaly density, and run-to-failure bias (Wu and Keogh, 2021). To mitigate these challenges, some scientists started to utilize newly synthesized datasets for their evaluations, allowing for a more fair comparison of TAD methods (Lai et al., 2021b; Zhang et al., 2019a; Ren et al., 2019; Feremans et al., 2020). In some cases, the development of new real-world datasets is also explored. WADI (Ahmed et al., 2017) and SWaT (Goh et al., 2017) are two prominent real-world datasets collected from sensors in a water distribution and treatment testbed, respectively. SMD (Su et al., 2019) is also another dataset containing data from server machines. In these datasets, training data is anomaly-free, and ground truth labels in the test data are recorded from pre-planned system changes. Although these datasets are widely used in TAD, it should be noted that anomaly-free training data aligns more with novelty detection branch than the anomaly detection. A collection of latest public datasets from various domains can also be observed in the works by Paparrizos et al. (2022) and Zamanzadeh Darban et al. (2024). Although a wide range of domains such as health (Moody and Mark, 2001), spacecraft telemetry data (Hundman et al., 2018), human activity recognition (Roggen et al., 2010; Bachlin et al., 2009), as well as traffic and transport (Ihler et al., 2006; Ahmad et al., 2017) can be observed in their collections, they have supplied their collections with several other synthetic datasets to provide a more comprehensive benchmarking opportunity. All these findings emphasize the lack of contributions to TAD with new *real-world* datasets. The recent

[1] https://github.com/imaantaheri/AnoLT

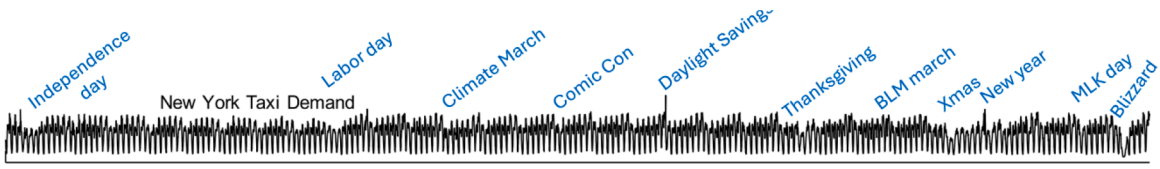


Fig. 1. An overview of a sample traffic benchmark dataset and its ground truth labels in TAD studies. Data shows six months of traffic demand in New York City (Ahmad et al., 2017; Nakamura et al., 2023).

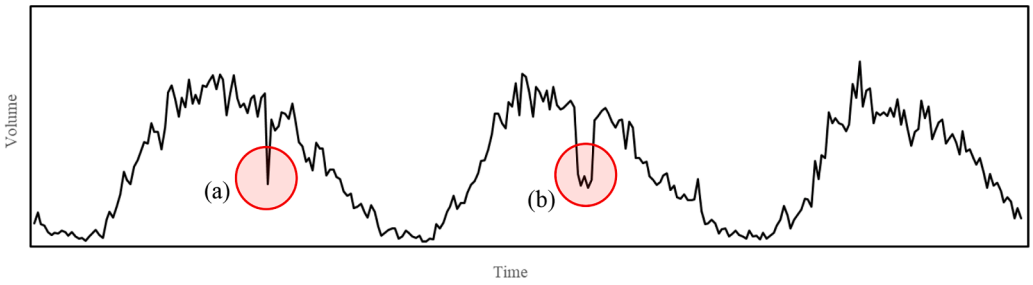


Fig. 2. Unexpected (a) point and (b) subsequence anomalies detected in traffic volume data on typical weekdays, with no specific events linked to these patterns.

success of Large Language Models (LLMs) (Touvron et al., 2023) also strongly underscores that access to ample and high-quality data can likewise result in significant advancements.

Transport and traffic-related datasets have been also utilized in many general TAD studies. NYC taxi demand (Ahmad et al., 2017), Dodgers dataset (Ihler et al., 2006), CalIt2 dataset (Hutchins, 2006), and Metro dataset (Hogue, 2019) are the most prominent examples. However, these datasets are limited in data size and also anomaly type. Additionally, ground truth labels in these datasets are only based on preexisting knowledge of events such as public holidays and planned network congestion (Nakamura et al., 2023). A sample of such data and its ground truth labels is demonstrated in Fig. 1. It can be observed that the annotated anomalies are only related to events that typically disrupt the whole daily pattern of the time series. However, upon closer inspection of traffic time series data, as shown in Fig. 2, other anomalies can clearly be identified during typical days of the week, which cannot be linked to predictable events. From a practical standpoint, it is more crucial to detect and annotate these types of anomalies with specific start and end points, rather than focusing only on long-lasting subsequences, as doing so would offer more opportunities to traffic planners and modelers to leverage the data. This fine-grained perspective will be ignored if we only rely on publicly recorded events in traffic datasets.

2.2. *Traffic/transport time series anomaly detection*

The detection of anomalous patterns in urban traffic data has long been a central focus of research. While some studies addressed anomaly detection in microscopic traffic data (Zarei Yazd et al., 2022; Sarteshnizi et al., 2022; Kazemeini et al., 2022), many others have concentrated on identifying anomalies, outliers, or events in macroscopic traffic time series data, mainly collected from fixed loop detectors embedded in road surfaces. Table 1 provides a detailed overview of recent studies in the latter domain, which is closely aligned with our focus. Based on the table, existing research can be categorized into two main branches: one focused on traffic incident detection, and the other on data-driven anomaly detection, which investigates data inconsistencies or irregularities in a more exploratory and generalized manner.

Incident detection studies aim to raise a flag when the normal flow of a road segment, usually a highway or freeway, is disrupted as a result of incidents. A variety of methods have been explored in this context, ranging from classical robust thresholding techniques (Balke et al., 1996; Chakraborty et al., 2019) to advanced deep learning architectures (Duan et al., 2024; Coursey et al., 2024). Since the objective in this category is to accurately detect incidents while minimizing false alarms, high resolution traffic data (sometimes at the lane level), with a strong emphasis on upstream and downstream traffic “speed”, is often employed (Chakraborty et al., 2019; Coursey et al., 2024). In line with this focus, incident reports are widely used as ground truth data in the evaluation pipelines of these studies (Kalair and Connaughton, 2021); however, the limitations of such records have also been acknowledged, as they often lack temporal and spatial accuracy due to the unexpected nature of incidents. Patterns in traffic speed data have also been recently used to manually or automatically refine incomplete incident labels, leveraging time-space diagrams or applying empirical thresholds to deviations from historical speed profiles (Coursey et al., 2024; Duan et al., 2024; Grigorev et al., 2023).

On the other side, new efforts are also emerging that exclusively focus on *unsupervised* data-driven anomaly detection in traffic time series data. Rather than confining attention to events or incidents, these approaches aim to identify inconsistencies that deviate from dominant patterns, following a more exploratory process that allows the data to reveal anomalies *on its own*. This is why, as shown in Table 1, they mostly generate synthetic anomalies and produce generalizable anomaly scores, as their detection objectives

Table 1
Overview of recent research papers focusing on traffic anomaly and incident detection[a].

Ref	Focus	Method	Input Data	GT Data	Output	Baseline Methods[b]
Chakraborty et al. (2019)	Incident detection	Robust Speed Thresholds	Spatiotemporal speed data	Verified incident reports	Anomaly labels (*ICT*)	SND (Balke et al., 1996); MAD (Hampel, 1974); IQR (Pearson, 2005)
Kalair and Connaughton (2021)	Anomaly/ Incident detection	KDE deviation	Volume and speed data (non-spatial)	Incident report & link travel times	Anomaly scores & labels (*ICT*)	Robust SND (Chakraborty et al., 2019); McMaster algorithm (Persaud and Hall, 1989)
Grigorev et al. (2023)	Incident detection & duration prediction	Profile Deviation & Segmentation	Speed data (non-spatial)	Incident reports & expert-labeled speed disruptions	Anomaly scores & labels (*ET*)	Comparison made with labeled disruptions (duration prediction benchmarked with state-of-the-art)
Duan et al. (2024)	Early incident detection	G-Trans trained by label-denoised data	Probe speed & data density levels, incident reports, weather data	Slowdown-based denoised incident reports	Anomaly labels (*ST*)	*Unsupervised methods include*: OOD (predictive) (Bahdanau et al., 2014); Mean Profile Deviation (Grigorev et al., 2023)
Coursey et al. (2024)	Incident detection	Evaluating autoencoder structures	Spatiotemporal volume, speed, & occupancy	Incident reports & expert-labeled speed disruptions	Anomaly scores & labels (*ICT*)	Benchmarked 6 different reconstruction-based deep autoencoders
Zhang et al. (2018)	Data-driven anomaly detection	Cross-region similarity check + OC-SVM	Spatiotemporal traffic data	Synthetic anomalies & Hit Rate (HR) with event reports	Anomaly scores	EE (Estimator, 1999); Deviation from nearby regions' mean; Likelihood Ratio Test (Doraiswamy et al., 2014)
Wang and Sun (2021)	Data-driven anomaly detection	Low-Rank Tensor Auto-regression	Spatiotemporal traffic data	Synthetic anomalies	Network anomaly scores	Three Factorization-based and Kalman filter-based methods (Ting et al., 2017)
Deng et al. (2022)	Data-driven anomaly detection	Spatio-Temporal Graph Conv-Adversarial Network (STGAN)	Spatiotemporal traffic data	Hit Rate (HR) with incident & event reports	Anomaly scores	iForest (Liu et al., 2008); LOF (Breunig et al., 2000); EE (Estimator, 1999); Vector Auto-Regression (Rousseeuw and Leroy, 2005); CIAS&AIAS (Zhang et al., 2018); ST-Decompn (Zhang et al., 2019b); & two advanced predictive methods
Sarteshnizi et al. (2023)	Data-driven anomaly detection	TP-FDS & Fast-ABOD	Volume and speed data (non-spatial)	Expert-labeled flow-density deviations	Anomaly scores	19 unsupervised ML and DL anomaly detectors evaluated *within* the 2D TP-FDS space.
He et al. (2023)	Data-driven anomaly detection	Automatic Reinforcement Learning	Volume data (non-spatial)	Synthetic subsequence anomalies	Anomaly labels	Telemanom (Hundman et al., 2018); DeepAnt (Munir et al., 2018); Breakout Detection (Vallis et al., 2014)
Ren et al. (2024)	Data-driven anomaly detection	Mirror Temporal Graph Autoencoder	Spatiotemporal traffic data	Incident reports & Synthetic anomalies	Anomaly scores	Historical average deviation, along with 3 other autoencoder-based, and 10 GNN-based deep learning models
The present study	Data-driven anomaly detection	Si-TDD (discord-based)	Volume data (non-spatial)	Expert-labeled time series inconsistencies	Anomaly scores	More than 20 baselines from TAD studies (*univariate*) and relevant previous traffic AD works

[a] GT: Ground Truth, *ICT*: Incident-based Calibrated Thresholds used, *ET*: Empirical Thresholds used, *ST*: Supervised Training.
[b] For detailed descriptions of individual baselines, the corresponding cited papers can be used.

go beyond certain incident or event labels (Wang and Sun, 2021; Ren et al., 2024). This field is under active development, with ongoing investigations from different viewpoints continuing to expand the body of knowledge. For instance, Zhang et al. (2018) proposed a windowed similarity-based method to identify traffic regions that show minimal differences relative to one another, and used a OC-SVM to detect disruptions in these similarity patterns. Wang and Sun (2021) introduced a method for anomaly detection in spatiotemporal traffic data using Low-Rank Tensor Auto-regression, validating their approach through synthetic datasets and visualizations of some real-world data scenarios. Alongside conducting various experiments, their work primarily emphasized detecting abrupt changes as anomalies within traffic data, placing greater focus on theoretically capturing spatiotemporal dependencies. Deng et al. (2022) and Ren et al. (2024) also proposed deep graph-based architectures for anomaly detection in spatiotemporal traffic data. For evaluation, they utilized incident reports and recorded social events along with PeMS and NYC taxi datasets. To derive final performance metrics, they applied strategies such as label-overlapping, synthetic anomaly injection, and hit-rate (recall)-based evaluation to address the lack of comprehensive ground truth labels. He et al. (2023) also suggested an anomaly detection technique for traffic flow (volume) time series data that employs Reinforcement Learning (RL) and Long Short-Term Memory (LSTM) networks, requiring no prior knowledge during calibration. Their approach only focused on subsequence anomalies, excluding point anomalies, and was validated using synthetic data alongside a number of real-world cases. Sarteshnizi et al. (2023) also concentrated on identifying deviations in temporal positioning of flow and density samples, arguing that these deviations effectively capture both point

and subsequence anomalies. Additionally, they benchmarked various anomaly detection methods for detecting deviations in their TP-FDS data space with a fully monitored and expert-labeled volume and speed dataset of ten different locations.

According to Table 1, these works usually rely on synthesized data with injected anomalies or perform targeted comparisons using reported incidents or public events. However, it has been argued that synthetic data is produced to support specific claims (Paparrizos et al., 2022), and incident/event reports are also shown to not completely reflect all types of data anomalies recorded in traffic data (Sarteshnizi et al., 2023). Additionally, solutions to alleviate the deficiencies of incomplete label sets have also proven to be problematic in recent TAD studies (Kim et al., 2022). Another noteworthy aspect in Table 1 is also the selection of baselines in comparative studies. Although many papers claim to address anomaly detection, and the ground truth labels indicate a broader scope beyond incident reports, there is limited consistency in baseline selection, with a strong emphasis on advanced deep learning methods and comparatively less attention to classical and univariate approaches (Ren et al., 2024; He et al., 2023). As discussed, classical methods have recently demonstrated both competitiveness and robustness in TAD with refined evaluation pipelines (Mejri et al., 2024), yet this is not fully reflected or investigated here.

This study differs from the above works in two main ways. First, in contrast to many anomaly detection studies that ultimately rely on event reports or synthetic data for evaluation, our study adopts a different perceptive. We investigate this challenge by enabling the visual annotation of all pattern-driven anomaly types directly based on the data and using these labels to evaluate existing solutions. This data-centric approach, which identifies anomalies based solely on observed time series inconsistencies, has not been explored in prior work, despite its potential to support a wide range of downstream exploratory analyses and automated applications. Second, unlike incident detection studies, we focus on anomaly detection in urban traffic volume data. While benchmark datasets and experiments exist for freeway incident detection (Coursey et al., 2024), comparable studies for volume data from stop-line loop detectors on arterials are lacking. This is particularly important as data collected from signal control systems, such as SCATS, is inherently different from freeway data and continues to grow (Dakic and Stevanovic, 2018; Yazdani et al., 2023). The spatial placement of these detectors within the network also differs from that on freeways, as they are located at the stop-line of incoming intersection approaches, making existing spatiotemporal speed-based labeling techniques inapplicable in this context.

3. Method

In this section, we first study the taxonomy of anomaly types in arbitrary time series data, and then discuss their appearance in traffic volume data. Afterwards, we elaborate on our proposed framework for real-world traffic data anomaly labeling. We then dive into our formulation of anomaly detection, and discuss our proposed Si-TDD solution.

3.1. Anomalies in time series data

In many recent studies, different types of anomalies/outliers are defined as contextual, point, and collective anomalies (Pang et al., 2021; Li et al., 2020a). However, Lai et al. (2021b) recently argued that this categorization does not precisely reflect all anomaly types in time series data, and they proposed new definitions based on spectral analysis (Shumway et al., 2000) to offer a more inclusive classification of anomaly types. We elaborate on this new taxonomy here and will later revisit it in the context of traffic volume data. It is important to note that this taxonomy reflects the findings of a recent fundamental work in TAD. Therefore, we aim to base both our labeling framework and methodology development on it to ensure that all state-of-the-art anomaly types are considered.

According to Lai et al. (2021b), a multivariate general time series data, named $\mathbf{X}$, can be represented by the following equation:

$$\mathbf{X}(t) = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_t, \dots, \mathbf{x}_T\} \tag{1}$$

where $\mathbf{x}_t$ is an N-dimensional vector recorded at time step t, and $t \in \{1, 2, \dots, T\}$ with T being the length of the series. For univariate time series data, we have $N = 1$, and therefore, each $\mathbf{x}_t$ in Eq. (1) denotes a scalar value. The main target in TAD is to assign a binary value or *label* to each observation ($\mathbf{x}_t$) recorded in the series, indicating whether the recorded data at time t is an anomaly or not. From a spectral analysis perspective, the pattern of a univariate $\mathbf{X}$ (named $\mathbf{S}$) can also be written as below:

$$\mathbf{X}(t) = \mathbf{S}(t, \omega) = \rho(2\pi\omega t) + \tau(t) \tag{2}$$

with ρ and τ functions being the *cyclic* and *trend* components of the series, and ω being the set of frequency indices of the cyclic behavior (number of cycles per unit of time). In this representation, the trend function shows the direction of the series, and the cyclic behavior can be shown by the sum of q different sinusoidal waves as follow:

$$\rho(2\pi\omega t) = \sum_{k=1}^{q} \rho_k(2\pi\omega_k t) \tag{3}$$

where

$$\rho_k(2\pi\omega_k t) = A_k cos(2\pi\omega_k t) + B_k sin(2\pi\omega_k t), \tag{4}$$

and in above equations, ω_k is the frequency index of each component, and A_k and B_k control the amplitude (maximum displacement) of the wave.

Considering both of the above formulations of time series data, Lai et al. (2021b) defined point-wise and pattern-wise anomalies as in Fig. 3. In point-wise anomalies, a specific point value of a series is considerably different than the expected value as follows:

$$|\mathbf{x}_t - \hat{\mathbf{x}}_t| > \delta(t) \tag{5}$$

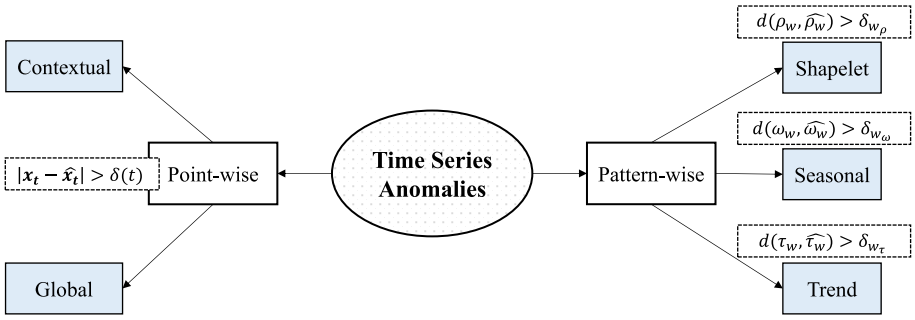


Fig. 3. An overview of the taxonomy of anomaly types in time series data, derived from the work by Lai et al. (2021b). The appearance of these anomaly types will be studied in traffic volume data.

where $\hat{\mathbf{x}}_t$ is the expected value of the point, and $\delta(t)$ is a time-sensitive discriminating threshold. Two sets of $\delta(t)$ can be applied to differentiate between *global* and *contextual* point-wise anomalies: one based on the surrounding points $(\mathbf{x}_{t-m}, \mathbf{x}_{t-m+1}, \dots, \mathbf{x}_{t+m-1}, \mathbf{x}_{t+m})$ for contextual anomalies, and the other one based on the whole values of the series $(\mathbf{X})$ for global ones.

In addition this type of anomalies, subsequences of data with anomalous behavior can also be labeled as pattern-wise anomalies. If we define a window or a subsequence, denoted as w, with the length of l as follows:

$$w(t) = [t, t + l] \tag{6}$$

and assume that the local spectral components of this window is extracted and is shown by $\mathbf{S}_{w(t)}$, this subsequence can be regarded as an anomaly when the following condition is true:

$$d(\boldsymbol{\varphi}_w, \hat{\boldsymbol{\varphi}}_w) > \delta_w \tag{7}$$

where d is a dissimilarity measurement function, δ_w is a threshold, and $\boldsymbol{\varphi}_w$ is one of the elements included in $\mathbf{S}_{w(t)}$ that can be equal to either ρ_w, ω_w, or τ_w. In case of dissimilarity between the cyclic function (ρ_w) of the real and expected subsequence, a *shapelet* anomaly is observed since the basic shape of the time series is disturbed. Seasonal anomaly also happens when Eq. (7) is satisfied with $(\varphi_w, \hat{\varphi}_w) = (\omega_w, \hat{\omega}_w)$. In this case, only the frequency of the series is abnormal and is not aligned with expectation (not the shape). And lastly, trend anomaly is referred to when the trend component of the series (τ_w) changes drastically while the other components remain the same afterwards. In other words, a subsequence of data leading to a permanent shift in the series can be called a trend anomaly.

Based on the taxonomy outlined above, it can be understood that anomaly types are defined solely based on the univariate progression of data. It is worth mentioning that this approach is not exclusive to the work by Lai et al. (2021b) and is prevalent in numerous prior works (Blázquez-García et al., 2021; Zamanzadeh Darban et al., 2024). To the best of our knowledge, no anomaly type described in the TAD literature has been yet proven to be theoretically *undetectable* when focusing solely on the univariate record of data (Sarteshnizi et al., 2023). The underlying reason for adopting multivariate modeling in previous works is mostly to improve the ability to identify such anomalies (Tian et al., 2023). However, by definition, anomalies are fundamentally expected to be evident from the univariate progression of the data itself. This is also the key rationale behind our focus on univariate modeling of traffic data in this work. While researchers in various domains of TAD achieve significant success with univariate solutions and consider them a promising approach for even handling multidimensional data (Nakamura et al., 2023; Lai et al., 2021b; Liu and Paparrizos, 2024; Paparrizos et al., 2022), their capacity has yet to be investigated in the context of urban traffic. In addition to its own applications, univariate modeling needs to be accomplished before any spatiotemporal modeling of data so that the added value of advanced modeling (if any) becomes more evident and discussable upon comparison. Despite the significance of this perspective, discussions on the nature and occurrence of anomalies in traffic time series data, as well as efforts to develop detection methods aligned with these underlying characteristics, remain lacking.

3.2. Anomalies in traffic volume data

Similar to the formulation of general time series data, a whole traffic volume data, which reflects the total number of vehicles passing a specific section of the road within a certain period of time, can also be shown with a spectral analysis point of view as below:

$$\mathbf{S}_v(t, \omega) = \rho_v(2\pi\omega t) + \tau_v(t) \tag{8}$$

with v referring to volume data, and all other notations defined previously. Two major factors in traffic volume/flow data need to be discussed according to its intrinsic features:

- First, the trend component in traffic data can be considered negligible $(\tau_v(t) \approx 0)$. The justification behind this assumption is that the number of vehicles passing a road section cannot exceed the capacity of the road, nor can it be less than zero. Therefore, the whole range of data is expected to vary between zero and the road capacity.

- Second, it is also proven that time-of-the-day and day-of-the-week features are always two substantial and ongoing external factors that significantly contribute to the cyclic pattern (ρ_v) of data in every location.

Therefore, based on above characteristics, Eq. (8) can be revisited as follow:

$$\mathbf{S}_v(t,\omega) = \rho_v^d + \rho_v^a + \rho_v^r \tag{9}$$

where ρ_v^d is a collection of periodic components that reflect a daily and constant cyclic behavior, ρ_v^a is a collection of components that adjusts ρ_v^d to include the effects of the day-of-the-week (such as the difference between workdays and weekends), and ρ_v^r is a substitute for the remaining components. ρ_v^r can include the effects of other factors such as seasonal or month-of-the-year patterns. In this work, we assume that the changes reflected by ρ_v^r are not intense, and therefore, $\rho_v^r \approx 0$. This is also a realistic assumption because of two main reasons:

1. These seasonal (or long-term) patterns of traffic data are so subtle that the previous literature utilizes complex methods to uncover them (Taheri Sarteshnizi et al., 2023).
2. It is also not theoretically possible to observe *considerable* variations in traffic demand in a road section for a long time (such as one month). Travelers usually adjust their travel time avoiding congested roads, and this results in a long-term balance in any unaltered road network (Sarteshnizi et al., 2023; Chiu et al., 2011).

Considering the above formulation and justifications, it can be then claimed that traffic volume data is mainly affected by a repetitive weekly pattern and it can be represented as follows:

$$\mathbf{S}_v(t,\omega) = \rho_v^d + \rho_v^a \tag{10}$$

According to the primary definitions of anomaly types, a number of conclusions can be stated here: (1) Trend anomaly cannot be observed in traffic volume data as $\tau_v(t) \approx 0$, (2) seasonal anomalies (abnormal frequency changes in data) cannot be defined for traffic volume data as well, since the collection of w in ρ_v^d and ρ_v^a is only a function of the constant daily and weekly behavior of data, and (3) global or contextual point and shapelet anomalies are the only types that can be observed in traffic data as there is no guarantee for a data point or subsequence to always follow the expected behavior represented by ρ_v^d and ρ_v^a in Eq. (10).

Based on above points, if a decent reflection of ρ_v^d and ρ_v^a is provided, any deviation from this pattern needs to be marked as either a point or shapelet (subsequence) anomaly. It should be also noted that shapelet anomalies in volume data may last for a few minutes to even a number of days (such as when a sensor failure happens). A long subsequence anomaly in traffic data should not be misinterpreted as a trend anomaly. A trend anomaly is when the whole series shift into a new range, and this range is then considered as a new normal representation of the system. Since long sensor failures are expected to end at one point, the whole data should be labeled as anomaly until when it returns into its normal range.

3.3. Volume data labeling framework

In this work, we assume that a certain amount of historical data (around 4 to 6 months) from any location is given, and the *majority* of this data (at least more than 50%) follows the normal behavior. Considering these prerequisites, we elaborate on our framework in the rest of this part.

Let us assume that the whole collected volume data from a specific location ($\mathbf{L}$) is as follows:

$$\mathbf{C_L} = \left\{ C_i^j \mid i \in \{1,2,.,7\}, j \in \{0,1,2,\dots,N\} \right\} \tag{11}$$

where C_i^j is the jth set of data points (or volume profile) collected on the ith day of the week. For instance, C_1^1 is the data of the first ($j = 1$) Monday ($i = 1$), and C_1^2 is the data of the second ($j = 2$) Monday ($i = 1$) within the whole historical data archive. Any C_i^j can also be defined as below:

$$C_i^j = \left\{ c_{(i,j)}^1, c_{(i,j)}^2, \dots, c_{(i,j)}^m, \dots c_{(i,j)}^M \right\} \tag{12}$$

with $c_{(i,j)}^m$ being the data point collected at the mth time step during the day. M is also always a fixed value demonstrating the number of points collected within a day, and its value is determined by the data collection time interval. For 15 min data resolution, as an example, we have $M = 96$.

Given the above formulation, we designed a labeling platform (as demonstrated in Fig. 4) and its characteristics (indicated by numbers on the figure) is described as below:

1. In our framework, data of each location ($\mathbf{C_L}$) is analyzed separately since anomalies should be evident in traffic data, given the periodic behavior of only that specific location (discussed before).
2. In each view of data, a specific and targeted C_i^j (named TC_i^j) is shown to the inspector along with other profiles having the same value of i (day-of-the-week). This provides the possibility of comparing TC_i^j with its own counterparts (profiles possessing the same day-of-the-week feature).
3. TC_i^j is shown with a different and noticeable color compared to the others as it is the only data part demanding inspection and labeling in each view. Other profiles with the same i as TC_i^j are shown in a unique and light color to allow easier comparison.
4. The inspector is also allowed to click on any data point from the TC_i^j (named $tc_{(i,j)}^m$) that seem to be anomalous, considering the pattern of other related profiles (having the same i value).

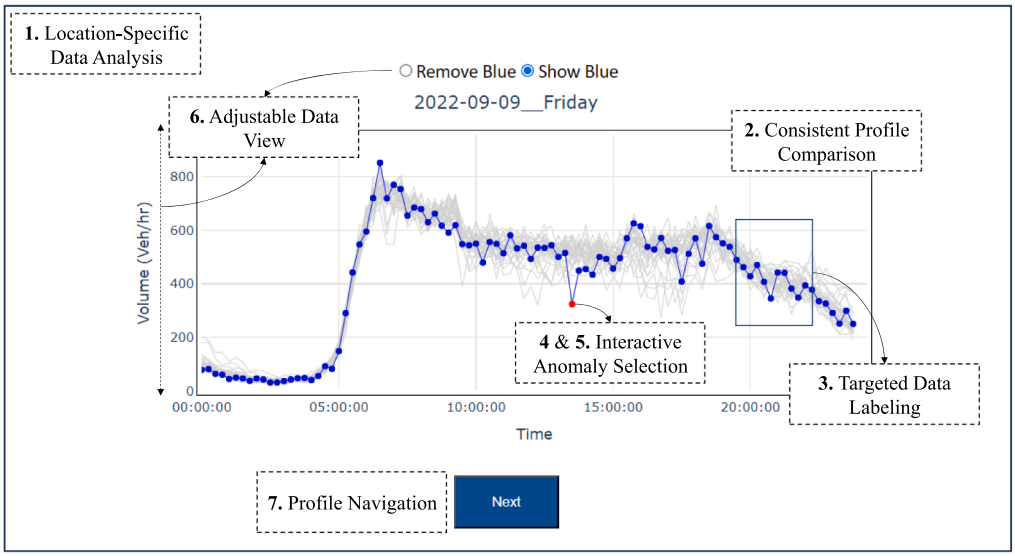


Fig. 4. A sample plot of data in our proposed data labeling framework. Each number is a reference to the characteristics described in the main text.

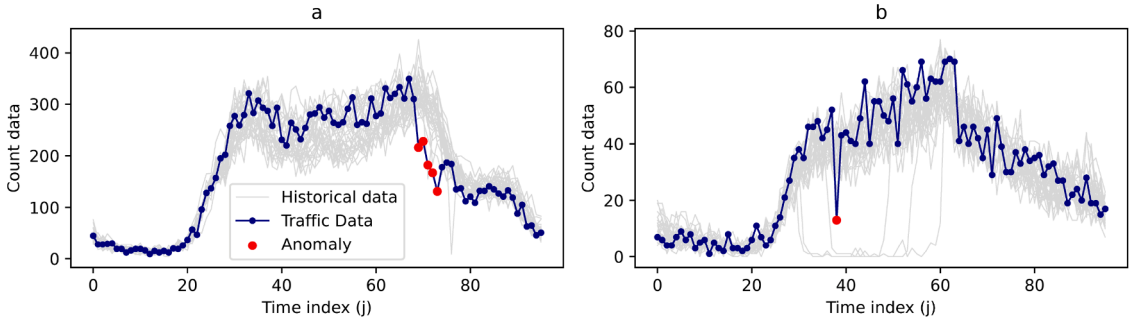


Fig. 5. Two traffic data profiles along with their counterparts. Shapelet (a) and point (b) anomalies selected by the inspector are shown in red.

5. All $tc^{m}_{(i,j)}$ in each view are clickable, and upon selection, they change color from blue to red indicating that the data has been selected by the inspector as an anomaly. Inspectors are also given the option to deselect those points by clicking again, restoring them to their normal state if required.
6. Scrolling over the Y-axis (volume values) as well as the possibility to hide the TC^{j}_{i} are provided. Therefore, in the case of a global anomaly or a subtle deviation, inspectors can zoom in and out to select data with greater precision and analysis.
7. A "Next" button is also displayed below each data view, allowing inspectors to move to the next TC^{j}_{i} once its labeling is complete. All TC^{j}_{i} profiles are arranged to be presented to the inspector sequentially until the whole data for the location is fully monitored.

Fig. 5 illustrates two additional views of data in our labeling platform, where TC^{j}_{i} samples, along with anomalies selected by an inspector, can be observed. When data monitoring is completed, all the TC^{j}_{i} from a location, a new label set is generated, as shown below, similar to the traffic data formulation:

$$\mathbf{G_L} = \left\{ G^{j}_{i} \mid i \in \{1, 2, ., 7\}, j \in \{0, 1, 2, \ldots, N\} \right\} \tag{13}$$

with G^{j}_{i} defined as:

$$G^{j}_{i} = \left\{ g^{1}_{(i,j)}, g^{2}_{(i,j)}, \ldots, g^{m}_{(i,j)}, \ldots g^{M}_{(i,j)} \right\}. \tag{14}$$

Similarly, G^{j}_{i} represents the set of binary labels collected for the jth profile of the ith day-of-the-week. $g^{m}_{(i,j)}$ is also a binary value indicating whether the data point collected at the mth time step of the day is normal or anomalous ($0 = normal$ and $1 = anomaly$). $\mathbf{G_L}$ will then be used in experimental studies to compare the performance of different anomaly detectors with traffic volume data.

As shown in Fig. 5, monitoring the data in this way provides the inspector with estimates of both ρ^{d}_{v} and ρ^{a}_{v}. This is ensured by displaying the data in daily profiles (ρ^{d}_{v}), where comparisons are only made with others recorded on the same day of the week (ρ^{a}_{v}). As a result, both types of anomalies-point and shapelet-can be efficiently identified during visual inspection. Inspectors are

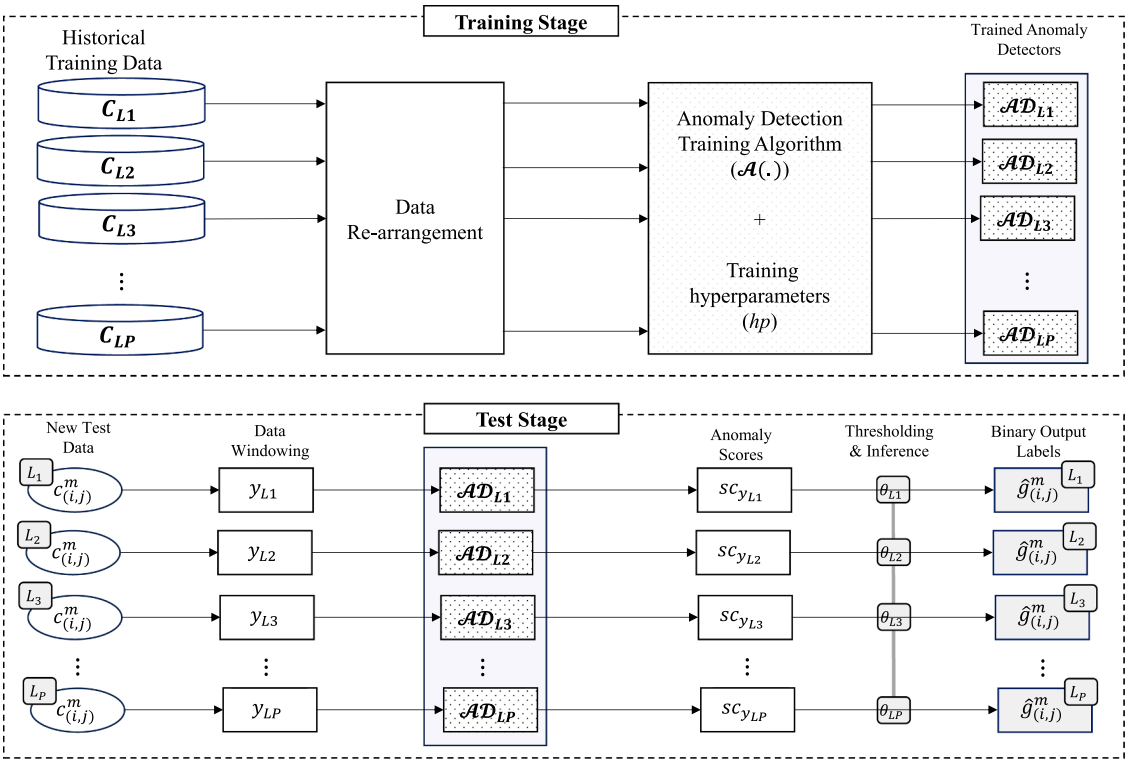


Fig. 6. An overview of the training and testing procedures employed in this study for anomaly detection in traffic volume data.

continuously given the opportunity to compare a data point with its surrounding points (Eq. (5)) and to compare a subsequence of data with the expected cyclic function of the series (Eq. (7)). Another important aspect of this framework is also its ability to enhance the precision of the judgment process compared to analyzing raw traffic time series data, a method commonly used in the literature (Fig. 1). For example, the shapelet anomaly in Fig. 5(a) might go unnoticed without the ability to compare it with similar historical data. These anomalies are not tied to specific events but instead represent non-recurrent and unpredictable situations, emphasizing the importance of thorough monitoring in effective evaluation.

3.4. Traffic anomaly detection: problem formulation

The overall procedure employed for traffic time series anomaly detection is illustrated in Fig. 6. As discussed earlier, this work focuses on univariate anomaly detection. Therefore, separate training and testing stages are included, where the data from each location is modeled individually using the same unique training algorithm. In the training stage, given the historical data of each location ($\mathbf{C}_{Lp}$ with p referring to the location), a dedicated model is trained using the ($\mathcal{A}(.)$) algorithm along with its required hyper-parameters (hp). Then, location-specific anomaly detectors ($\mathcal{AD}_{Lp}(.)$) are obtained as below:

$$\mathcal{AD}_{Lp}(.) = \mathcal{A}(\mathbf{C}_{Lp}, hp) \tag{15}$$

During the test stage, when new data sample(s) arrive, they are supplied along with a portion of recent data in a window (y_{Lp}) to their relevant models. This concept may also be used during the training stage (in data re-arrangement part in Fig. 6). Anomaly scores ($sc_{y_{Lp}}$) are then calculated based on y_{Lp} by:

$$sc_{y_{Lp}} = \mathcal{AD}_{Lp}(y_{Lp}). \tag{16}$$

Scores from Eq. (16), are then expected to be significantly higher when an anomaly is present in y_{Lp}. Depending on the model structure, anomaly scores of individual samples ($sc^m_{(i,j)}$) are then interpreted from these window scores ($sc_{y_{Lp}}$). In the final step, the sample scores ($sc^m_{(i,j)}$) are converted into binary labels, using the following threshold function:

$$\hat{g}^m_{(i,j)} = \begin{cases} 0 & \text{if } sc^m_{(i,j)} < \theta_{Lp}, \\ 1 & \text{if } sc^m_{(i,j)} \geq \theta_{Lp} \end{cases} \tag{17}$$

where θ_{Lp} denotes the *user-defined* and constant threshold for each location. Benefiting from the collection of $\hat{\mathbf{G}}$, a point-by-point comparison can be performed with the achieved ground truth labels (Eq. (14)) to compute performance metrics such as AUC-ROC and F1 score (Lai et al., 2021b).

In this formulation, the main objective is to identify an optimal Anomaly Detector ($\mathcal{A}^*$) capable of mapping raw data ($c^m_{(i,j)}$) to anomaly scores ($sc^m_{(i,j)}$), a task that is not trivial and has proven to be challenging across many applications (Blázquez-García et al., 2021). $\mathcal{A}^*$ can be represented as follows:

$$\mathcal{A}^* = \arg\min_{\mathcal{A}} \mathcal{L}\left(\mathbf{G}, \hat{\mathbf{G}}\right) \tag{18}$$

with $\mathcal{L}(.)$ being the loss function quantifying the discrepancies between the real ($\mathbf{G}$) and predicted ($\hat{\mathbf{G}}$) ground-truth labels. In this work, we focus on investigating Eq. (18) in the context of *fine-grained* traffic volume anomaly detection. We also assume that the set of θ_{Lp} in the final stage is determined by the user. This is the case because (1) it is a widely adopted approach in both traffic and broader anomaly detection literature (Table 1), allowing users to select thresholds based on their preferences, and (2) extracting optimal anomaly scores from raw time-series data is a non-trivial task in itself, requiring dedicated studies to address it. The impact of threshold settings and their applicability is best examined in future works once reasonable scores have been successfully extracted.

3.5. Traffic anomaly detection: methodology

To elaborate on our proposed method ($\mathcal{A}^*$), we begin with an overview of the procedures behind existing TAD solutions. We then focus on time series discord, a concept that forms the foundation of our approach. Finally, we introduce our Si-TDD method and discuss how it enhances discords discovery solutions to address the unique requirements of traffic volume data.

3.5.1. TAD solutions

Various TAD methods can be applied to the formulation above as a substitute for $\mathcal{A}(.)$. These solutions are broadly categorized into two distinct groups: *prediction-based* and *similarity-based* approaches (Blázquez-García et al., 2021). The main difference between these methods lies in how they frame the problem and interpret it. Eq. (19) represents the y_{Lp} window (with the length of $\lambda + 1$) at two consecutive time steps (m and $m+1$), where $c^m_{(i,j)}$ and $c^{m+1}_{(i,j)}$ denote the latest data at location L_p:

$$y_{Lp} = \begin{cases} \{c^{m-\lambda}_{(i,j)}, \dots, c^{m-1}_{(i,j)}, c^m_{(i,j)}\} & \text{at } t = m, \\ \{c^{m-\lambda+1}_{(i,j)}, \dots, c^m_{(i,j)}, c^{m+1}_{(i,j)}\} & \text{at } t = m+1. \end{cases} \tag{19}$$

Prediction-based methods typically use recent data in y_{Lp} to predict the current value of $c^m_{(i,j)}$, utilizing the trained model. Then, the anomaly score $sc^m_{(i,j)}$ (or $sc^{m+1}_{(i,j)}$) is directly calculated by comparing the estimated value ($\hat{c}^m_{(i,j)}$) with the collected data, as shown below (same can be derived for $m+1$):

$$sc^m_{(i,j)} = sc^m_{y_{Lp}} = \left| c^m_{(i,j)} - \hat{c}^m_{(i,j)} \right| \tag{20}$$

However, in *similarity-based* methods, the time-window anomaly score ($sc_{y_{Lp}}$) is first computed benefiting from the degree of dissimilarity between the y_{Lp} window and the collection of all other historical time-windows (Y^{hist}_{Lp}), as in the following equation:

$$sc^m_{y_{Lp}} = d\left(y^m_{Lp}, Y^{hist}_{Lp}\right) \tag{21}$$

with $d(.)$ being a dissimilarity measurement function. This function can range from a simple Euclidean distance to a complex dissimilarity boundary defined in an advanced data representation system. Then, the final sample score ($sc^m_{(i,j)}$) is *extracted* from the window score $sc^m_{y_{Lp}}$ in a specific way, depending on the application and requirements of the user.

Since similarity-based methods provide anomaly scores for the entire input sequence ($sc^m_{y_{Lp}} \neq sc^m_{(i,j)}$), additional post-processing steps are often required in some cases to obtain the point-wise anomaly labels. Although this is specific to similarity-based approaches, it is worth noting that prediction-based methods are also trained to minimize prediction error, which may not directly align with the goal of accurate anomaly detection. As a result, their performance in identifying anomalies can also be suboptimal.

3.5.2. Time series discord for anomaly detection

Despite significant progress with deep learning approaches, recent comparative studies have shown that classical methods, mostly from the similarity-based group, are still state-of-the-art tools for TAD (Paparrizos et al., 2022; Lai et al., 2021b; Liu and Paparrizos, 2024). Discord discovery is one of the most widely studied solutions in TAD problems. Time series discord refers to a subsequence of time series data that is significantly dissimilar to all other subsequences of the same length within the series, with this dissimilarity often measured using the Euclidean distance function (Nakamura et al., 2023). To be specific, if we define a univariate time series as in Eq. (1), a subsequence of data with the length of λ and its latest data point recorded at time $q \in \{1, 2, 3, \dots, T\}$ (denoted by y^q_λ here) is regarded as the discord of the series if it satisfies the following condition:

$$ED\left(y^q_\lambda, y^{(nn,q)}_\lambda\right) > ED\left(y^o_\lambda, y^{(nn,o)}_\lambda\right) \quad \forall o \in \{1, 2, \dots, T\},\ o \neq q. \tag{22}$$

In this condition, $ED(.)$ represents the Euclidean distance measurement function, y^o_λ denotes other subsequences of data with the same length λ at time o, and $y^{(nn,q)}_\lambda$ and $y^{(nn,o)}_\lambda$ are the nearest neighbor subsequences to y^q_λ and y^o_λ, respectively.

Based on above, to determine whether a subsequence is a time series discord, its distance to the nearest neighbor must be compared with that of other subsequences. Although the definition centers around the top first discord (y^q_λ), this concept is extensively leveraged to rank and detect all anomalies in time series data. To achieve this, one way is to compute pairwise distance measurements between

all subsequences within the series and then consider the discord scores (distances to nearest neighbors) as anomaly scores. Algorithm 1 presents this default approach of transforming the concept into a practical anomaly score generator. In this method, with no training required, the discord scores for all subsequences of length λ in a time series $\mathbf{X}(t)$ are computed, with a computational complexity of $O(n^2)$ (all subsequences analyzed against each other). These scores, as discussed in Section 3.5.1, are associated with data *subsequences* and may need to be processed through application-based criteria to classify anomalies in time series data.

Algorithm 1 Time series anomaly detection with discord discovery.

Require: Time series data: $\mathbf{X}(t)$, subsequence length: λ
Ensure: Ranked subsequences with discord scores
Extract subsequences of length λ from time series data $\mathbf{X}$:
for $q = \lambda$ to T **do**
  $y^q \leftarrow \mathbf{X}[q - \lambda + 1, q]$ ▷ Extract subsequence ending at time q
end for
$discord_scores \leftarrow []$ ▷ Initialize list to store discord scores
for each subsequence y^q **do**
  $distances \leftarrow []$ ▷ Initialize list to store distances
  for each subsequence y^o where $o \neq q$ **do**
    $d_{qo} \leftarrow ED(y^q, y^o)$ ▷ Compute Euclidean distance between subsequences
    append d_{qo} to $distances$
  end for
  $discord_score_q \leftarrow \max(distances)$ ▷ Find the maximum distance (discord score)
  append $discord_score_q$ to $discord_scores$
end for
return $(discord_score_q, y^q)$ ▷ Return data subsequences along with their scores

In time series discords discovery, selecting the optimal window size is often considered as a significant challenge. Fig. 7 elaborates on this in the context of traffic volume data. In this figure, the positions of representative time-windows of three sizes ($\lambda \approx$ one day, one hour, and a few hours) rolling over volume data can be found (y^{τ_1}, y^{τ_2}, and y^{τ_3}). In the case of anomalous *daily profiles* or *long-duration* anomalies, as in Fig. 7(a), a large window size of approximately one day is highly effective. A high discord score is produced by comparing the anomalous day with its nearest neighbor, a data window similar to the one next to it. Detection of such anomalies (e.g., public holidays or large events) have been addressed many times before (Fig. 1). However, for more subtle *point* or *short-duration* subsequence anomalies, as in Fig. 7(b) and (c), there are situations that may not be precisely reflected. For instance, in Fig. 7(c), it is observed that a large or medium window ($\lambda >$ a few hours) may not accurately differentiate between point anomalies that are in close proximity. This is because the normal segment between them, which might be shorter than the window length, does not get the chance to be recognized as an isolated normal part. Additionally, smaller windows, as in Fig. 7(b), may lack the necessary context, causing low discord scores for some true anomalies. Despite these challenges, variants of discord discovery methods have remained state-of-the-art in several TAD applications (Lin et al., 2005; Yeh et al., 2024).

3.5.3. Step-isolated traffic discords discovery (Si-TDD)

Si-TDD follows the path of discord discovery methods, extending the application of this intuitive and practical approach to traffic volume data. The core idea of Si-TDD is to transform the window-based discord discovery method outlined above into a step-wise process. This adaptation not only aligns more closely with the unique characteristics of traffic volume data but also addresses challenges related to both complexity and window-size selection. Recall our main formulation in Sections 3.4 and 3.5.1. In Si-TDD; two major modifications are applied to tailor the concept. First, we always set $\lambda = 1$ in Si-TDD, meaning that each test data point $c^r_{(a,b)}$ at location Lp is the only included element in y_{Lp} window to be labeled at each time interval:

$$y_{Lp} = \left\{c^r_{(a,b)}\right\}. \tag{23}$$

Secondly, rather than calculating the discord scores of y_{Lp} against all subsequences, as in Eqs. (21) and (22), we compare the data exclusively with its $\mathcal{K}$th nearest neighbor within a *designated subset* of historical data. This subset consists of data points collected at the same time of day and on the same day of the week as the incoming data. In other words, we take advantage of the cyclic temporal features of volume data (discussed earlier) to facilitate and improve discord discovery by only focusing on the most relevant parts of the data each time.

Fig. 8 provides an overview of the whole anomaly detection procedure using Si-TDD. At the beginning, a specific collection of historical data (Y^{hist}_{Lp}) is first constructed by grouping $c^m_{(i,j)}$ data points that share the same i (day-of-the-week) and m (time-of-the-day) values, as shown below:

$$Y^{hist}_{Lp} = \left\{\mathbf{y}^{(i,m)}_{Lp} \mid i \in \{1, 2, \dots, 7\}; m \in \{1, 2, \dots, M\}\right\}, \tag{24}$$

where each $\mathbf{y}^{(i,m)}_{Lp}$ collection includes the following elements:

$$\mathbf{y}^{(i,m)}_{Lp} = \left\{c^m_{(i,j)} \mid j \in \{1, 2, \dots, N\}\right\} \tag{25}$$

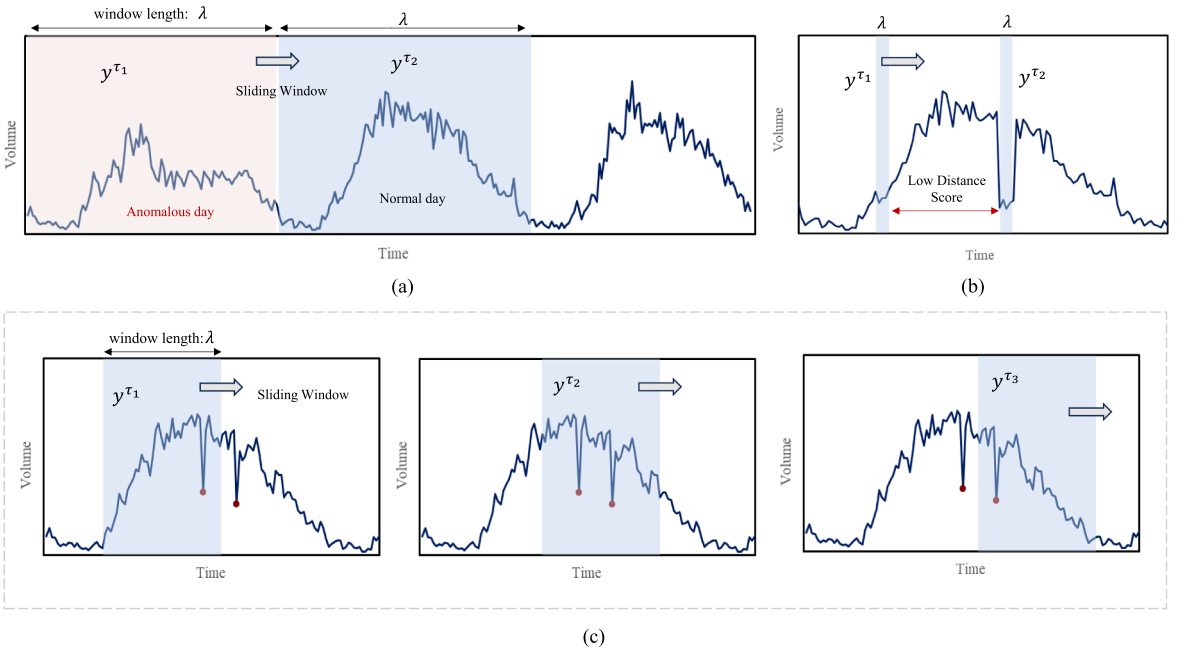


Fig. 7. Traffic volume anomaly appearance in sliding windows of discord discovery based on time-window duration: (a) large: $\lambda \approx$ one day, (b) small: $\lambda \approx$ one hour, and (c) medium: $\lambda \approx$ a few hours.

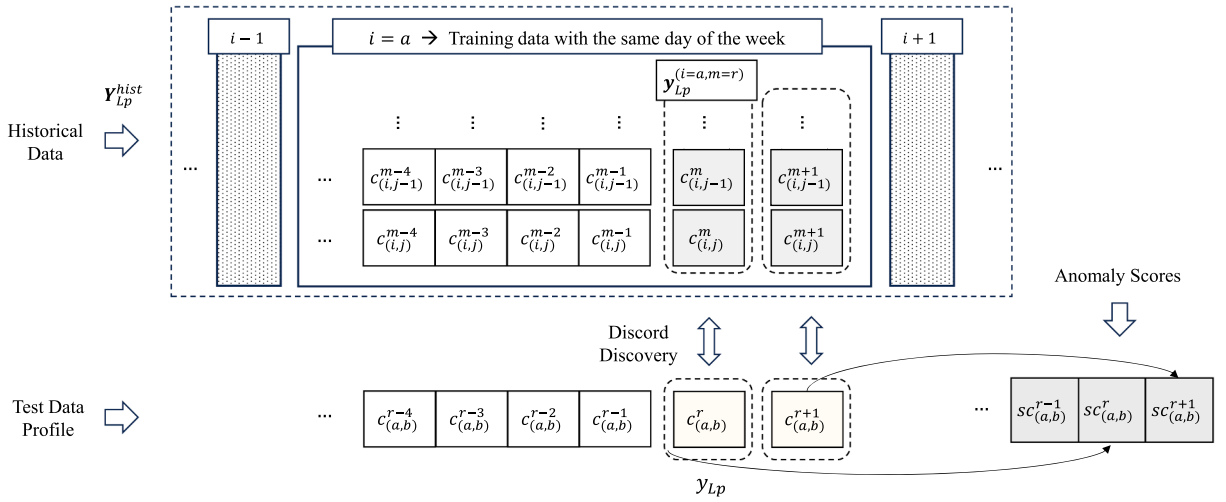


Fig. 8. A demonstration of anomaly score calculation procedure with the Si-TDD method.

with all notations defined previously. Then, anomaly score of y_{Lp}, which is also equivalent to the anomaly score of $c^r_{(a,b)}$, is computed as the Euclidean distance between y_{Lp} and its $\mathcal{K}$th nearest neighbor in $\mathbf{y}^{(i,m)}_{Lp}$, where $i = a$ and $m = r$ (denoted by $j = nn$):

$$sc^r_{(a,b)} = sc_{y_{Lp}} = ED(c^r_{(a,b)}, c^{m=r}_{(i=a,nn)}). \tag{26}$$

The reason for selecting the $\mathcal{K}$th nearest neighbor here is to also avoid using a similar anomaly from the training data as the basis for distance calculation. According to this formulation, anomaly scores in Si-TDD are linked to the distance between the test samples and their homogeneous historical data. Therefore, they increases as samples deviate from the expected range, with larger scores reflecting greater deviations. The complete algorithm for anomaly detection based on Si-TDD can be found in Algorithm 2. This algorithm transforms the discord discovery problem for traffic volume data into a more straightforward yet precise procedure, requiring the configuration of only a single hyperparameter ($\mathcal{K}$).

A number of key points regarding Si-TDD need to be mentioned. According to the methods and baselines listed in Table 1, it can be inferred that the potential of step-wise *discord discovery* in traffic volume data has received little to no attention in prior studies, despite the intuitive nature of its core idea. Similar profile deviation approaches, such as SND-like variants (Kalair and Connaughton, 2021) and mean profile deviation methods (Grigorev et al., 2023; Ren et al., 2024), have been widely used as classical methods to benchmark against event-based ground truth labels. However, Si-TDD differs meaningfully from these methods, offering a robust perspective that enables effective score generation for all anomaly types. First, Si-TDD remains unaffected by the presence of large deviations in the training data, as such anomalies are well known to distort conventional average-based profiles (Chakraborty et al., 2019). Second, unlike other robust SND-like detection approaches, it does not assume any underlying distribution or central tendency for traffic data at each time step, as empirical evidence suggests that such assumptions do not hold true at all times (Kalair and Connaughton, 2021). Furthermore, we will show that the anomaly scores produced by Si-TDD create a more accurate and reliable basis for threshold selection in binary anomaly detection than existing deviation-based and parametric methods, even when optimized parameters are used along with those methods. Relevant experimental evaluations are provided in the following section to support these statements, and demonstrate the effectiveness of Si-TDD.

Algorithm 2 Anomaly detection with step-isolated traffic discords discovery (Si-TDD).

Require: historical data: $\mathbf{C}_{Lp}$, newly arrived data: $c^{m=r}_{(i=a,j=b)}$, $\mathcal{K}$th nearest neighbor to consider: $\mathcal{K}$.
Ensure: New data anomaly score: $sc^{r}_{(a,b)}$.

$\mathbf{y}^{(a,r)}_{Lp} \Leftarrow []$ ▷ Initialize list to store homogeneous samples
for all data points ($c^{m}_{(i,j)}$) in $\mathbf{C}_{Lp}$ collection **do**
  if $i = a$ and $m = r$ **then**
    append $c^{m}_{(i,j)}$ sample to $\mathbf{y}^{(a,r)}_{Lp}$ list.
  end if
end for
$dist \Leftarrow []$ ▷ Initialize list to store distance to homogeneous data
for all $c^{r}_{(a,j)}$ samples in $\mathbf{y}^{(a,r)}_{Lp}$ list **do**
  $d_j \Leftarrow$ Euclidean distance of $c^{r}_{(a,j)}$ to $c^{r}_{(a,b)}$.
  append d_j to the $dist$ list.
end for
$sorted_dist \Leftarrow$ descending order of values in $dist$ list. ▷ Sort the list of distance to candidates
$sc^{q}_{(a,b)} \Leftarrow \mathcal{K}$th item in $sorted_dist$ list. ▷ Find the $\mathcal{K}$th nearest neighbor
return $sc^{q}_{(a,b)}$ ▷ Return the new data anomaly score

4. Experimental results

4.1. AnoLT dataset

To establish the AnoLT (Anomaly Labeled Traffic) dataset using our labeling platform, we benefited from SCATS (Sydney Coordinated Adaptive Traffic System) stop-bar loop detectors of 147 road sections (sites) spatially connected together within Melbourne, Australia.[2] These sites are part of two different hypothetical networks, as in Fig. 9(a), where each point represents an intersection equipped with loop detectors. It can be observed from Fig. 9(b), that these detectors are typically installed per lane at each incoming direction of intersections as a result of the SCATS system, primarily used for traffic control systems and signals management (Yazdani et al., 2023). The AnoLT dataset encompasses data from all links within the targeted networks, with volume observations aggregated from lane-level to link-level at each intersection. The dataset contains seven months of data for each of the specified networks, recorded at a 15 min temporal resolution. Data for the Dandenong Road network spans from February to August 2019, and for the AIMES network it extends from May to December 2022. To ensure data reliability, we conducted a primary data monitoring process across all sites, confirming that the dataset predominantly consisted of normal operational data. Links exhibiting consecutive periods of faulty data-characterized by negative or extremely high values-over an extended duration (e.g. three to four months) were excluded from the dataset.

We developed an interactive web application using Dash from Plotly (Hossain et al., 2019) to design and implement our labeling platform (as in Fig. 4), enabling the collection of ground truth data. To start the labeling procedure, two experts were familiarized with the labeling framework system and instructed on the various types of anomalies that may be present in traffic volume data. Prior to the main labeling experiment, a trial session was initially conducted in which the inspectors were provided with volume data not included in the final main dataset. During this session, inspectors were encouraged to collaboratively label the data, discuss their findings, and align their *standards* for anomaly labeling. In the main experiment, however, each inspector independently labeled the data without input from the other person to avoid any potential bias. The labeling task for each inspector was completed over

[2] SCATS traffic volume data for the state of Victoria in Australia is publicly accessible. Check out the following URL: https://discover.data.vic.gov.au/dataset/traffic-signal-volume-data/historical

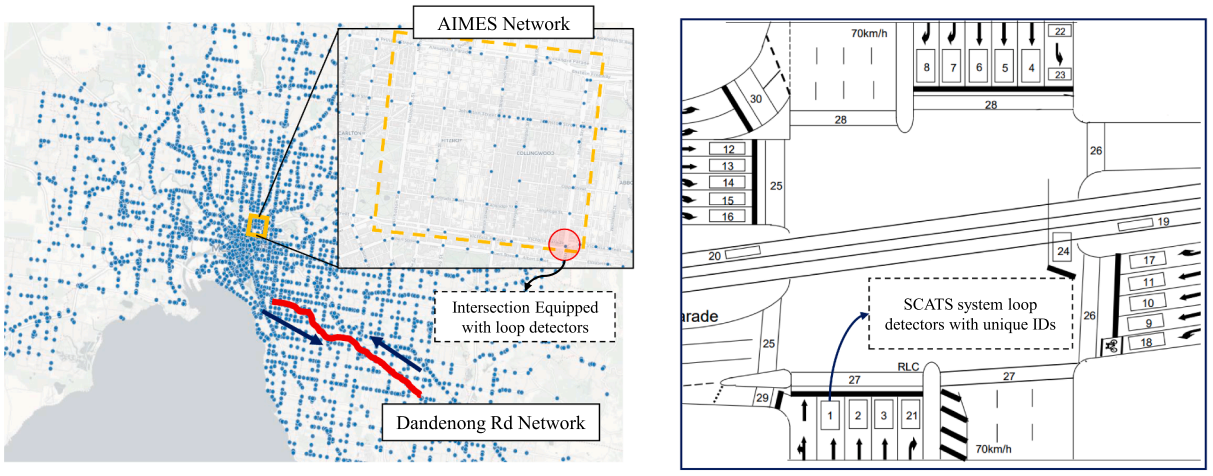


Fig. 9. Visualization of (a) the targeted road networks in Melbourne, Australia, and (b) a sample layout of loop detector placement at an intersection within the SCATS system. Volume data from the AIMES and Dandenong Road networks are utilized in this study.

a period of approximately three months, with daily work loads exceeding two hours. In the end, for each link $\mathbf{L}$ (or location), two different binary label sets ($\mathbf{G}_L^{\alpha}$ and $\mathbf{G}_L^{\beta}$) became available, indicating whether the inspector has recognized individual samples as an anomaly or not (as in Eqs. (13) and (14)). In addition, $\mathbf{G}_L^{\alpha\cap\beta}$ and $\mathbf{G}_L^{\alpha\cup\beta}$ were also generated for each location, resulting in four different sets of ground truth labels in the AnoLT dataset.

Table 2 summarizes the overall characteristics of the AnoLT dataset, which includes over 3 million monitored volume data points. The table also highlights key differences between the two label sets, noting that the α set contains a higher number of anomaly labels compared to the β. A more detailed analysis and comparison of the labels in AnoLT is presented in Table 3, which categorizes the labeled anomalies into four distinct types based on their durations (point anomaly, as well as short, medium, and large subsequence anomalies). The table indicates that while the average anomaly ratios are 3.20% in α and 2.76% in β, these values vary considerably across different locations. In some cases, it exceeds 10%, whereas in others, it falls below 1%. Notably, short shapelet anomalies are also the most frequently occurring category across the whole dataset.

The level of consistency, or agreement, between the two label sets, reported in Table 2, requires further attention and discussion. We measured this consistency, also referred to as inter-rater reliability (Saal et al., 1980), using the F1 score metric, with its average value being reported as 0.78. F1 has been shown to be a suitable and rigorous metric for TAD evaluation when no systematic adjustment is required between the data and the ground truth labels (Kim et al., 2022). Since the ground truth labels in AnoLT are not expected to exhibit any misalignment due to our point-wise collection approach, the F1 score can be directly calculated as shown below and will be used in this section to compare labels from different sources:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{27}$$

where Precision is defined as:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \tag{28}$$

and Recall is defined as:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}. \tag{29}$$

Table 2 indicates that an average 78% level of alignment exists between the collected labels in AnoLT. This is a relatively high value, as F1 scores closer to one represent stronger consistency. Detailed distribution of this cross-comparison across the labels derived for all locations in AnoLT is depicted in Fig. 10. It is observed that this metric ranges from 0.45 to 0.96, with the majority of samples or locations ($>90\%$) achieving values above 0.6. Based on this observed level of agreement, we concluded that having two distinct label sets in our dataset is sufficient for conducting a comprehensive comparative analysis across various methods. It will be also shown that many state-of-the-art methods, in their original form, face challenges in achieving an even comparable level of performance, while the Si-TDD method offers a significantly more reasonable alignment with these label sets.

In addition to the statistical evaluation, we also randomly visualized the data alongside the label sets in over 200 cases, as demonstrated in Fig. 11, to further explore the root causes of differences in more detail. We found that the selection of *boundary points* was the primary factor contributing to the majority of differences between the two label sets. This was particularly expected and also inevitable as the labelers were instructed to pay close attention to “subtle” point and shapelet anomalies, such as the second

Table 2
An overview of the overall characteristics of the AnoLT dataset.

Metric	First label set (α)	Second label set (β)
Total time series labeled	147	147
Total number of observations	3,125,856	3,125,856
Time series duration per location ≈	7 months	7 months
Time series collection resolution	15 min	15 min
Total number of anomalies identified	98,476	85,684
Total number of conflicting labels	25,303	12,511
Percentage of average anomalies per time series	3.2%	2.76%
Average level of agreement (measured by F1 score)	0.78 (std: 0.12)	

Table 3
Basic statistics related to the share of different anomaly types in the AnoLT dataset (Percentages represent their associated value *per location* in each label set).

Anomaly Type	Ratio in α			Ratio in β		
	Min	Max	Mean	Min	Max	Mean
Point anomalies (%)	0.13	3.15	0.93	0.09	1.63	0.60
Short shapelet anomalies (%, 2-8 points)	0.30	3.77	1.23	0.29	2.65	1.04
Medium shapelet anomalies (%, 8-32 points)	0.00	2.50	0.60	0.00	2.98	0.65
Long shapelet anomalies (%, >32 points)	0.00	6.87	0.43	0.00	6.69	0.46
All anomalies (%)	0.69	10.28	3.20	0.70	11.01	2.76

Note: Shapelet (subsequence) anomalies are categorized based on their length: *short* (2-8 points, lasting up to 2 h), *medium* (8-32 points, 2-8 h), and *long* (> 32 points or > 8 h).

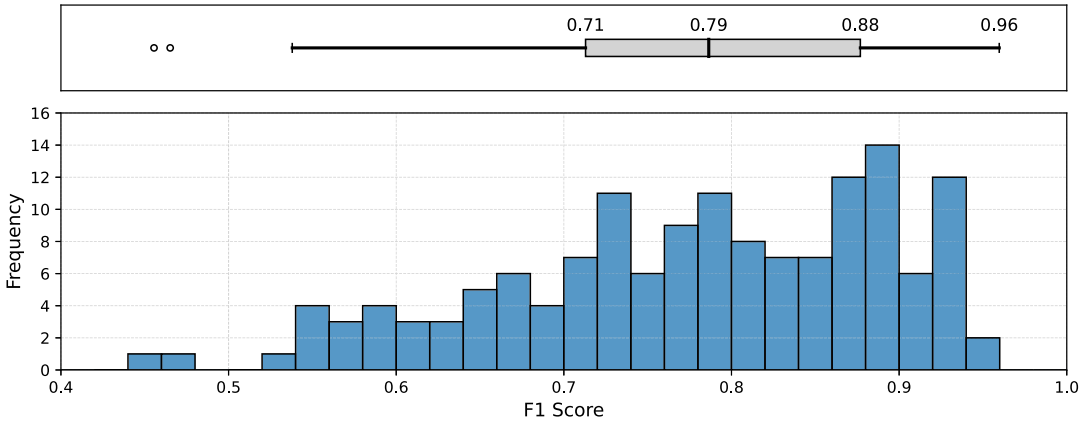


Fig. 10. Histogram of F1-scores observed across different locations, comparing the labels provided by the inspectors of the AnoLT dataset (inter-rater reliability).

anomaly in Fig. 11(a) or the first one in Fig. 11(e), and to maintain their standards consistently throughout the entire experiment. This phenomenon is also observed at the beginning or end of notable subsequence anomalies, as shown in Fig. 11(b) or (g), and in singular points where they potentially indicate the beginning of a major continuous anomaly, but the data has returned to normal behavior afterward (e.g., the third anomaly in Fig. 11(e)). Despite the existence of these differences resulting from the utilized level of sensitivity, a very high level of alignment was recorded for noticeable anomalies where an anomaly is clearly present within the data, as in Fig. 11(c).

4.2. Baselines

We included a diverse range of methods, spanning from classical to recently developed approaches, in our comparative study using the AnoLT dataset. *Similarity-based* solutions have received significant attention both historically and in recent studies, with numerous similarity measurement techniques proposed (Liu and Paparrizos, 2024). A part of this category contains methods originally designed for non-sequential data that have been adapted by the windowing approach to address TAD. **LOF** (Breunig et al., 2000), **OCSVM** (Schölkopf et al., 2001), **COF** (Tang et al., 2002), **PCA** (Shyu et al., 2003), **iForest** (Liu et al., 2008), **ABOD** (Kriegel et al., 2008), **HBOS** (Goldstein and Dengel, 2012), **LODA** (Pevnỳ, 2016), and **COPOD** (Li et al., 2020b) are selected from this branch of

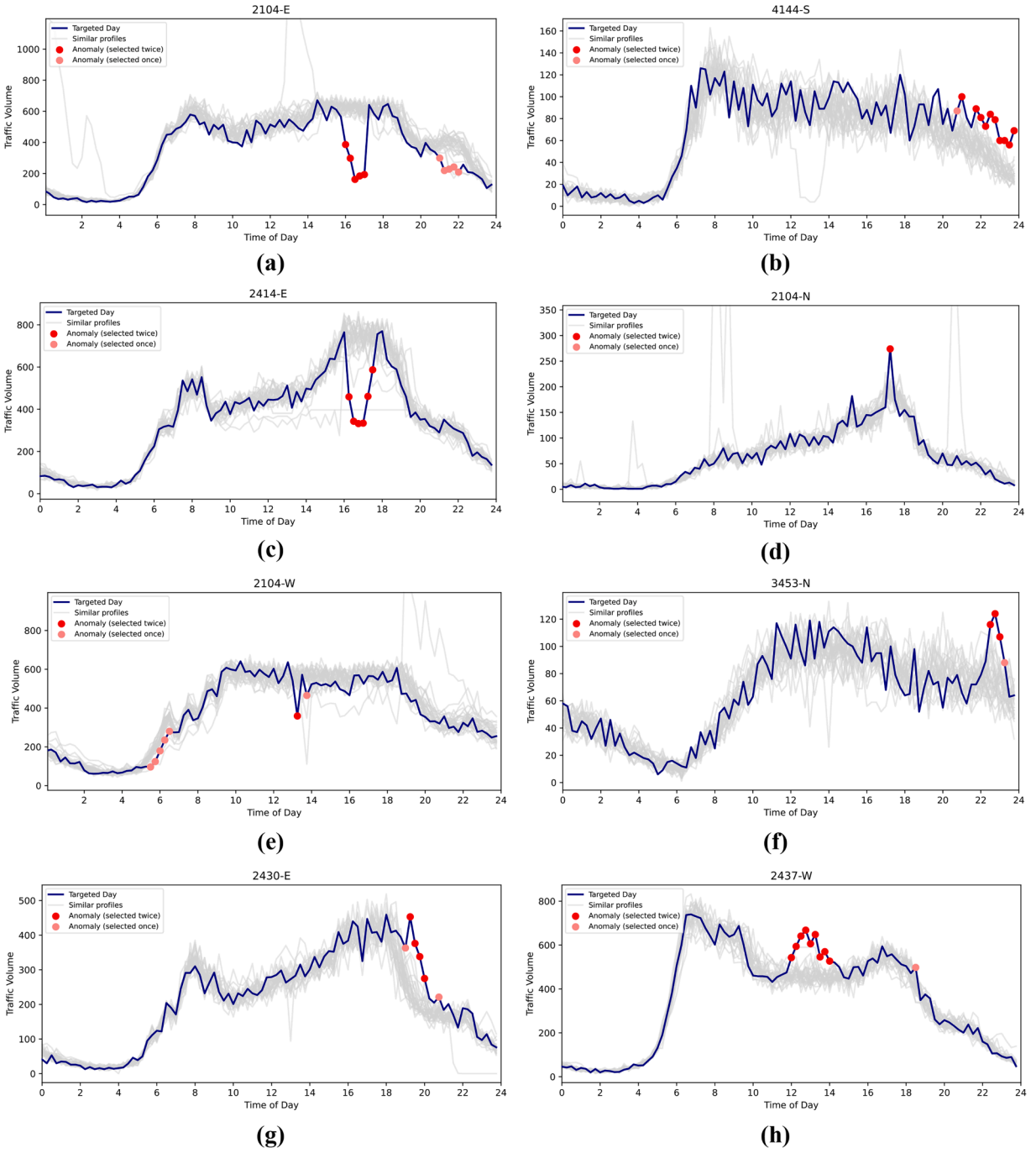


Fig. 11. Overview of anomalies recorded in the AnoLT dataset. Data selected by only one inspector is highlighted in light red color.

methods. We also added time series **Discord** (Angiulli and Pizzuti, 2002; Nakamura et al., 2023) and its variant, the STUMPY Matrix Profile (**MP**) method (Law, 2019), to this list as representatives of discord-based approaches. From the intersection of deep learning and similarity measurement, we picked Deep Autoencoder (**AE**), Variational Autoencoder (**VAE**) (Aggarwal, 2016), and generative adversarial networks, such as **SO-GAAL** and **MO-GAAL** (Liu et al., 2019). These methods typically estimate the underlying distribution of normal sequences and leverage it during the testing stage to identify anomalous data. We also included **STGAN** (Deng et al., 2022) and **Auto-RL** (He et al., 2023) in this group to reflect on domain-specific progress in traffic studies as well. In the category of *prediction-based* methods, we considered Auto Regression (**AR**) (Rousseeuw and Leroy, 2005), a classic and widely recognised algorithm that has demonstrated promising results in specific TAD applications. Additionally, we included **LSTM** (Bontemps et al., 2016), along

Table 4
Hyperparameter search spaces for baselines (**bold** values resulted in the highest performance).

Method	Hyperparameter	Search Space
MP	window_size and step_size	[3, 4, 5, …, **10**] and [**1**, 2, 3]
LOF	n_neighbors	5, 7, 10, 15, **20**
	window_size	3, 4, 5, 7, **10**
PCA	layers_structure	[3, 2], [3, 1], [4, 2], [4, 3], [5, 3], [5, 4], [5, 3], [5, 2], [7, 4], [7, 3], [7, 2], [10, 7], [10, 5], **[10, 3]**, [10, 2]
COF	n_neighbors	5, 7, 10, 15, **20**
	window_size	3, 4, 5, 7, **10**
ABOD	n_neighbors	5, 7, 10, 15, **20**
	window_size	3, 4, 5, 7, **10**
HBOS	n_bins and alpha	[5, 7, 10, 15, **20**] and [**0.1**, 0.2, 0.3, 0.5]
	tolerance and window_size	[**0.1**, 0.2, 0.3, 0.5] and [3, 4, 5, 7, **10**]
LODA	n_bins	5, 7, 10, 15, 20, 25, **30**
	n_random_cuts and window_size	[20, 50, 70, 100, 120, 150, **200**] and [3, 4, 5, 7, **10**]
iForest	bootstrap	**True**, False
	behavior and window_size	[‘old’, **‘new’**] and [3, 4, 5, 7, **10**]
Discord	window_size and method	[3, 4, 5, 6, 7, 8, 9, **10**] and [‘largest’, **‘mean’**, ‘median’]
	step_size and n_neighbors	[**1**] and [5, 7, 10, 15, **20**]
OCSVM	kernel and nu	[‘linear’, ‘poly’, **‘rbf’**, ‘sigmoid’] and [**0.4**, 0.5, 0.6]
	tolerance, shrinking, and window_size	[**0.001**] [**True**], and [3, 4, 5, 7, **10**]
COPOD	window_size	3, 4, 5, 7, **10**
AE	layers_structure	[3, 2, 1, 2, 3], [3, 2, 2, 2, 3], [4, 3, 2, 3, 4], [4, 3, 2, 1, 2, 3, 4], [5, 4, 3, 4, 5], [5, 4, 3, 2, 3, 4], [7, 5, 4, 3, 4, 5, 7], [7, 5, 3, 5, 7], [7, 4, 3, 2, 3, 4, 7], **[10, 7, 4, 7, 10]**, [10, 5, 4, 3, 4, 5, 10], [10, 5, 4, 3, 2, 3, 4, 5]
VAE	layers_structure	[3, 2, 1, 2, 3], [3, 2, 2, 3], [4, 3, 2, 3, 4], [4, 3, 3, 3, 4], [5, 4, 3, 3, 3, 4, 5], [5, 4, 3, 2, 3, 4, 5], [7, 5, 4, 3, 4, 5, 7], [7, 5, 4, 2, 4, 5, 7], [7, 4, 3, 4, 7], **[10, 7, 4, 4, 7, 10]**, [10, 5, 4, 3, 4, 5, 10], [10, 5, 4, 3, 2, 3, 4, 5, 10]
SO-G and MO-G	n_epochs and k for MO-G	**20** and **10**
	window_size	3, 4, 5, 7, **10**
AR	window_size and step_size	[3, 4, 5, 6, 7, 8, 9, **10**] and **1**
LSTM	hidden_dim and n_epochs	[**32**, 64] and **20**
	window_size and n_hidden_layer	[**3**, 5, 7, 10] and [**3**, 5, 10]
	min_anomaly_time	**1**, 2, 3, 4
DeepLog	hidden_dim	32, **64**
	n_epochs and dropout_rate	**20** and **0.1**
	window_size	3, 5, 7, **10**
	stacked_layers	2, 3, **4**
Telemanom	smoothing and n_prediction	**0.05** and **1**
	n_epochs and dropout_rate	**20** and **0.1**
	window_size	**3**, 5, 7, 10
	layers	[10, 10], [10, 10, 10], **[10, 10, 10, 10]**
STGAN	λ for anomaly score	[range 0 to **1**]
	recent_time, and num_feature	**1** h (default), and **1** (volume only)
	timestamp, and n_subgraph_nodes	**4** (15-min samples per hour), and **9** (default)
	time_feature	**31** (24 h + 7 days per week)
Auto-RL	state_len and window_smoothing	[**15**, 20, 25] and [**1**, 2, 3, ., 10]
	kde_timesteps_included	[isolated, ± **2 steps**]
Si-TDD	n_neighbors	1, 2, **3**, 4, 5, 6, 7, 8, 9, 10

with its recent advancements, **DeepLog** (Du et al., 2017) and **Telemanom** (Hundman et al., 2018), as learning-based methods for prediction-driven anomaly detection.

The methods described above were implemented using TODS (Lai et al., 2021a) and PyOD (Zhao et al., 2019), two state-of-the-art Python packages specifically designed for benchmarking anomaly detection methods. For STGAN, we also leveraged the spatial relationships of the road networks included in AnoLT, as required by the detection architecture, using the code implementation provided in the main paper (Deng et al., 2022). We also implemented Auto-RL based on the architecture described by He et al. (2023), as the original code was not publicly available. For each location, the dataset was divided into training (70%) and testing (30%) subsets, and evaluation was achieved by recording the highest point-wise F1 score of the test set derived from varying thresholds applied to anomaly scores. Hyperparameter tuning was also performed using a grid search over a predefined and reasonable search space, which also included the default values. These search spaces of baselines are outlined in Table 4, where the optimal hyperparameter combinations identified for the AnoLT dataset are highlighted in bold.

To further discuss the effectiveness of Si-TDD, we also exclusively evaluate its performance against existing profile-based methods. These methods, also known as SND-like approaches, follow the principles of similarity-based methods and stem from classical incident detection research. Although incident detection works have shifted toward upstream-downstream analyses on freeways to reduce false alarms (a capability enabled by adjacent loop detectors installed along the road or connected vehicle data), SND-like methods remain effective baselines for site-specific tasks, particularly in contexts lacking strong spatial connections between sites (Chakraborty et al., 2019; Kalair and Connaughton, 2021). These methods compute deviations at each time of day and day of the week, with different variants described below:

- **SND** (Balke et al., 1996): Computes the standard normal deviate, flagging anomalies when the observed value exceeds the mean by c times the standard deviation.
- **IQR** (Pearson, 2005): Flags anomalies when the observed value is less than $Q_1 - c \times \text{IQR}$ or greater than $Q_3 + c \times \text{IQR}$, where Q_1 and Q_3 denote the first and third quartiles, and $\text{IQR} = Q_3 - Q_1$.
- **SND-R** (Kalair and Connaughton, 2021): A robust variant of SND that uses the median instead of the mean and the inter-quartile range (IQR) instead of the standard deviation, flagging anomalies when the observed value exceeds the median by c times the IQR.
- **MAD** (Hampel, 1974): Uses the median and the mean absolute deviation (MAD) as the dispersion measure, flagging anomalies when the observed value exceeds the median by c times the MAD.
- **Med-D**: A robust adaptation of average profile distance methods (Grigorev et al., 2023; Ren et al., 2024) (as averages are sensitive to large outliers), using the absolute distance between the observed value and the median profile at each time of day and day of the week directly as the anomaly score.

For the relevant methods, the threshold scaling parameter c is fully optimized for each location by testing values in the range of *0.05* to *10*, with a step size of *0.001*, ensuring that the optimal performance in all cases falls within this range.

4.3. Comparative studies

Tables 5 and 6 present the performance metrics of all baselines evaluated against the ground truth labels in AnoLT (α, β, $\alpha \cap \beta$, and $\alpha \cup \beta$). Fig. 12 further complements these tables by visualizing the distribution of F1 scores across different locations for a number of selected methods, including the Si-TDD. The results clearly demonstrate that Si-TDD significantly outperforms other baselines in terms of anomaly score calculation, achieving an F1 score exceeding 67% and an AUC-ROC of 97%. This is also further supported by Fig. 12, which shows that Si-TDD achieves scores greater than 0.4 across various locations, underscoring its robust performance and reliability in all scenarios. In contrast, other state-of-the-art methods, from both prediction-based and similarity-based approaches, are shown to be considerably less effective when applied to traffic volume data, with the majority achieving average F1 scores of approximately 45% or lower. The distributions in Fig. 12 also highlight these low performances across different locations in comparison to Si-TDD.

Specific factors can be identified to explain the obtained results. For *similarity-based* TAD methods, the primary challenge lies in the unique cyclic behavior inherent in traffic volume data, which differs significantly from many other types of datasets. Methods in this category are capable of delivering satisfactory results in other contexts, as the cyclic behavior of data in these scenarios is often repeated multiple times within a short period. For example, in ECG data, a complete time series can be generated within just a few seconds, with anomalies typically manifesting as distinct abnormal patterns (Wu and Keogh, 2021). In such cases, any short-term subsequence y_{L_p} (as discussed before), even with a duration of just a few seconds, generally contains at least a few complete and meaningful cycles, making window-based similarity comparisons an effective solution. In traffic volume data, however, a complete cycle spans an entire day (24 h), meaning that window lengths ranging from 3 to 10 samples (equivalent to 45 to 150 min) fail to capture a distinguishable or meaningful pattern. As a result, low performance is expected when using similarity-based methods with traffic data structured in such window lengths. Although increasing the window length to one day offers a viable solution for detecting anomalous *daily profiles*, it is shown here to be unsuitable for detecting fine-grained (within-day) anomalies in traffic volume data (as discussed in Fig. 7 for the discord discovery method).

Although developed for traffic data, STGAN and Auto-RL did not also found to yield comparable detection outcomes in this category. For STGAN, it was observed that the proposed structure, particularly the generator and discriminator components, was unable to effectively learn the underlying behavior of traffic data in AnoLT, despite the full input of temporal and spatial features and hyperparameter adjustments. This is due to factors such as the availability of only a single volume series per location (rather than additional series like speed and occupancy) and differences in detector placement, data size, and characteristics compared to the original study. Prior work has also shown that advanced deep learning architectures, while achieving strong performance in specific settings, may not generalize well to other contexts (Lai et al., 2021b). Auto-RL has also been shown to detect substantial and prolonged subsequence deviations in experimental datasets containing at least 10% anomalies (He et al., 2023). In contrast, we found that AnoLT contains, on average, 2-3.5% anomalies (Table 2), with some major subsequence anomalies spanning only 5-6 samples (equivalent to 75-90 min given our 15 min collection interval). This presents additional challenges for effectively training the RL component within this method to differentiate between normal and abnormal samples in AnoLT.

Prediction-based methods do not suffer from the window length issue in similarity-based methods, however, their focus on reducing prediction error rather than detecting anomalies makes them less effective at identifying shapelet or subsequence anomalies. When faced with such anomalies, they often predict the next observation based on recent anomalous values to minimize prediction error, which mainly limits their applicability in TAD. Given that subsequence anomalies constitute a considerable portion of all anomalies, it is evident from Tables 5 and 6 that prediction-based methods are not sufficiently effective for AnoLT. This justification

Table 5
Accuracy metrics of TAD methods with the α and β label sets in AnoLT.

Name	Results based on α					Results based on β				
	AUC	Prec	Rec	F1	F-SD	AUC	Prec	Rec	F1	F-SD
MP	0.567	0.157	0.760	0.260	0.095	0.559	0.126	0.720	0.214	0.092
LOF	0.774	0.418	0.483	0.449	0.090	0.784	0.418	0.444	0.431	0.104
PCA	0.466	0.187	0.793	0.303	0.106	0.478	0.181	0.711	0.288	0.111
COF	0.723	0.348	0.479	0.403	0.075	0.732	0.342	0.435	0.383	0.089
ABOD	0.789	0.406	0.519	0.456	0.093	0.793	0.413	0.470	0.440	0.091
HBOS	0.624	0.211	0.663	0.320	0.108	0.634	0.192	0.585	0.289	0.110
LODA	0.614	0.233	0.639	0.342	0.103	0.630	0.230	0.575	0.329	0.109
iForest	0.588	0.212	0.635	0.317	0.098	0.605	0.186	0.559	0.279	0.108
Discord	0.774	0.398	0.527	0.454	0.093	0.778	0.406	0.478	0.439	0.093
OCSVM	0.634	0.259	0.555	0.353	0.094	0.646	0.257	0.490	0.337	0.110
COPOD	0.525	0.188	0.712	0.270	0.105	0.534	0.165	0.675	0.238	0.107
AE	0.457	0.189	0.840	0.309	0.105	0.471	0.180	0.779	0.293	0.111
VAE	0.450	0.190	0.826	0.309	0.103	0.461	0.179	0.771	0.290	0.109
SO-G	0.420	0.140	0.920	0.243	0.093	0.430	0.117	0.914	0.207	0.104
MO-G	0.412	0.139	0.925	0.241	0.093	0.423	0.116	0.911	0.206	0.104
STGAN	0.671	0.120	0.268	0.166	0.065	0.672	0.122	0.262	0.166	0.071
Auto-RL	0.729	0.238	0.289	0.261	0.139	0.743	0.231	0.321	0.268	0.143
AR	0.663	0.192	0.254	0.219	0.069	0.656	0.198	0.241	0.217	0.085
LSTM	0.620	0.071	0.400	0.120	0.064	0.612	0.058	0.390	0.101	0.064
DeepL	0.722	0.264	0.293	0.278	0.083	0.721	0.265	0.294	0.279	0.091
Telem	0.629	0.149	0.319	0.203	0.083	0.631	0.141	0.336	0.199	0.094
Si-TDD	**0.976**	0.684	0.668	**0.676**	0.075	**0.971**	0.680	0.649	**0.664**	0.086

Note: SO-G: SO-GAAL, MO-G: MO-GAAL, DeepL: DeepLog, Telem: Telemanom, AUC: AUC-ROC, Prec: Precision, Rec: Recall, F1: F1 score, F-SD: F1 score Standard Deviation.

is also mentioned before by findings of other TAD comparative studies (Liu and Paparrizos, 2024). Another noteworthy point also pertains specifically to deep learning approaches within this category, and potentially others. While advanced methods demonstrate exceptional performance in various self-supervised tasks, such as time series prediction, they are yet less favored for this applications. Given that TAD is an investigative, intuitive, and explanatory task, there is often a preference for straightforward and self-explanatory methods that align more closely with these characteristics (Nakamura et al., 2023). Additionally, challenges such as overfitting and highly parameterized architectures further limit the applicability of these methods to TAD applications (Wu and Keogh, 2021).

Fig. 13 is provided to illustrate the sensitivity of window size to the final accuracy of TAD methods. As noted, for similarity-based methods, mid-sized window lengths yield higher performance compared to smaller ones, as larger intervals represent more pattern-related information. Prediction-based methods, on the other hand, exhibit low sensitivity to window size, with their lower performance mainly attributed to factors such as the limitation in capturing context information and subsequence anomalies. It should be noted that window sizes larger than 10 points (2.5 h) were not evaluated, as assigning labels from large windows to individual points in similarity-based methods cannot be performed precisely and is incompatible with the nature of anomalies in AnoLT (Fig. 7). For prediction-based methods, the selected range also aligns with the standard practices in previous works (Qi et al., 2022).

The comparative results with profile-based (SND-like) approaches are also summarized in Tables 7 and 8. As noted earlier, to achieve these outcomes, we selected the optimal threshold value for each location based on the labels in AnoLT. For methods requiring the c coefficient, we employed a wide and fine-grained grid search (from 0.05 to 10 in increments of 0.001). For Med-D, thresholds were applied systematically based on the derived distance measures. The results show that Si-TDD continues to outperform these approaches, and this can be attributed to different factors. First, the underlying statistical assumptions of SND, IQR, SND-R, and MAD appear to be unsuitable for the traffic data in AnoLT. Even with extensive testing of finely incremented thresholds within these methods (c values), the resulting outcomes remain considerably inferior to those obtained by thresholding Si-TDD scores. Fig. 14 shows the density estimates of data collected at a specific location during different time intervals on a selected day of the week. It can be observed that the distribution of traffic volume data varies across time intervals and days, reflecting the inherent variability in normal traffic conditions. This challenges the assumption of stable central tendencies in SND-like methods, which assume that symmetric deviations from these central indicators represent anomalies.

Additionally, higher F1 outputs from Med-D and Si-TDD also indicate that using distance-based measurements to a reference point creates a more optimized search space for final threshold selection, leading to more accurate binary labels of anomalies. The superiority of Si-TDD over Med-D further lies in its ability to account for the normal variability present in the data. The importance of this matter is evident in Fig. 14, which shows that normal samples are not always symmetrically distributed around the median, leading to lower performance in locations where this asymmetry is more intense (as also denoted by the F-M column in Tables 7 and 8). This becomes even more critical in situations where bi-modal traffic patterns are present, with traffic variations in certain time intervals exhibiting two distinct regimes (Kalair and Connaughton, 2021). In such cases, the median value may be located near one

Table 6
Accuracy metrics of TAD methods with the $\alpha \cap \beta$ and $\alpha \cup \beta$ label sets in AnoLT.

Name	Results based on $\alpha \cap \beta$					Results based on $\alpha \cup \beta$				
	AUC	Prec	Rec	F1	F-SD	AUC	Prec	Rec	F1	F-SD
MP	0.560	0.107	0.672	0.184	0.077	0.567	0.178	0.777	0.289	0.091
LOF	0.796	0.419	0.423	0.421	0.112	0.765	0.415	0.513	0.459	0.084
PCA	0.486	0.181	0.653	0.283	0.111	0.459	0.195	0.841	0.316	0.100
COF	0.744	0.345	0.408	0.374	0.094	0.715	0.349	0.506	0.413	0.069
ABOD	0.807	0.423	0.445	0.433	0.098	0.780	0.410	0.546	0.469	0.088
HBOS	0.638	0.169	0.568	0.261	0.110	0.621	0.225	0.694	0.339	0.101
LODA	0.640	0.221	0.544	0.314	0.109	0.606	0.235	0.678	0.349	0.099
iForest	0.616	0.178	0.506	0.263	0.107	0.580	0.218	0.696	0.332	0.094
Discord	0.792	0.399	0.455	0.425	0.100	0.764	0.392	0.560	0.462	0.089
OCSVM	0.660	0.259	0.466	0.333	0.111	0.623	0.259	0.601	0.362	0.092
COPOD	0.541	0.149	0.637	0.212	0.106	0.519	0.207	0.739	0.294	0.098
AE	0.478	0.174	0.735	0.282	0.110	0.450	0.190	0.881	0.313	0.100
VAE	0.469	0.177	0.720	0.285	0.107	0.443	0.193	0.881	0.316	0.097
SO-G	0.430	0.095	0.910	0.172	0.088	0.419	0.161	0.921	0.274	0.094
MO-G	0.423	0.094	0.905	0.171	0.088	0.412	0.160	0.933	0.273	0.094
STGAN	0.681	0.121	0.241	0.161	0.076	0.664	0.123	0.293	0.173	0.060
Auto-RL	0.756	0.209	0.342	0.260	0.145	0.719	0.260	0.273	0.266	0.137
AR	0.659	0.205	0.216	0.211	0.088	0.661	0.192	0.272	0.225	0.068
LSTM	0.612	0.053	0.374	0.092	0.062	0.622	0.073	0.431	0.125	0.063
DeepL	0.728	0.274	0.277	0.275	0.095	0.718	0.253	0.311	0.279	0.081
Telem	0.637	0.149	0.289	0.197	0.098	0.624	0.136	0.362	0.198	0.080
Si-TDD	**0.978**	0.690	0.666	**0.678**	0.089	**0.971**	0.676	0.668	**0.672**	0.072

Note: SO-G: SO-GAAL, MO-G: MO-GAAL, DeepL: DeepLog, Telem: Telemanom, AUC: AUC-ROC, Prec: Precision, Rec: Recall, F1: F1 score, F-SD: F1 score Standard Deviation.

Table 7
Accuracy comparison with profile-based methods using the α and β label sets in AnoLT.

Name	Results based on α						Results based on β					
	AUC	Prec	Rec	F1	F-SD	F-M	AUC	Prec	Rec	F1	F-SD	F-M
SND	0.785	0.548	0.585	0.566	0.097	0.239	0.774	0.537	0.561	0.549	0.110	0.238
IQR	0.770	0.474	0.557	0.512	0.104	0.284	0.760	0.472	0.536	0.502	0.121	0.241
SND-R	0.762	0.487	0.543	0.513	0.104	0.294	0.753	0.477	0.521	0.498	0.122	0.246
MAD	0.764	0.463	0.548	0.502	0.104	0.255	0.754	0.467	0.525	0.494	0.121	0.259
Med-D	0.966	0.611	0.596	0.603	0.091	0.296	0.962	0.612	0.579	0.595	0.101	0.331
Si-TDD	**0.976**	0.684	0.668	**0.676**	0.075	**0.449**	**0.971**	0.680	0.649	**0.664**	0.086	**0.423**

Note: AUC: AUC-ROC, Prec: Precision, Rec: Recall, F1: F1 score, F-SD: F1 score Standard Deviation, F-M: Minimum F1 score observed.

Table 8
Accuracy comparison with profile-based methods using the $\alpha \cap \beta$ and $\alpha \cup \beta$ label sets in AnoLT.

Name	Results based on $\alpha \cap \beta$						Results based on $\alpha \cup \beta$					
	AUC	Prec	Rec	F1	F-SD	F-M	AUC	Prec	Rec	F1	F-SD	F-M
SND	0.776	0.545	0.564	0.554	0.112	0.240	0.780	0.558	0.576	0.567	0.094	0.250
IQR	0.759	0.472	0.531	0.499	0.123	0.241	0.773	0.482	0.568	0.521	0.102	0.305
SND-R	0.753	0.475	0.519	0.496	0.124	0.248	0.770	0.485	0.561	0.520	0.102	0.305
MAD	0.750	0.472	0.513	0.492	0.123	0.248	0.774	0.461	0.574	0.511	0.102	0.275
Med-D	0.970	0.626	0.590	0.607	0.107	0.255	0.961	0.604	0.592	0.598	0.085	0.373
Si-TDD	**0.978**	0.690	0.666	**0.678**	0.089	**0.432**	**0.971**	0.676	0.668	**0.672**	0.072	**0.444**

Note: AUC: AUC-ROC, Prec: Precision, Rec: Recall, F1: F1 score, F-SD: F1 score Standard Deviation, F-M: Minimum F1 score observed.

regime, causing normal observations from the other to be incorrectly flagged as anomalies. Measuring distances to neighbors, as in Si-TDD, better accommodates these situations by capturing local dispersion and the potential inherent clustering within the data.

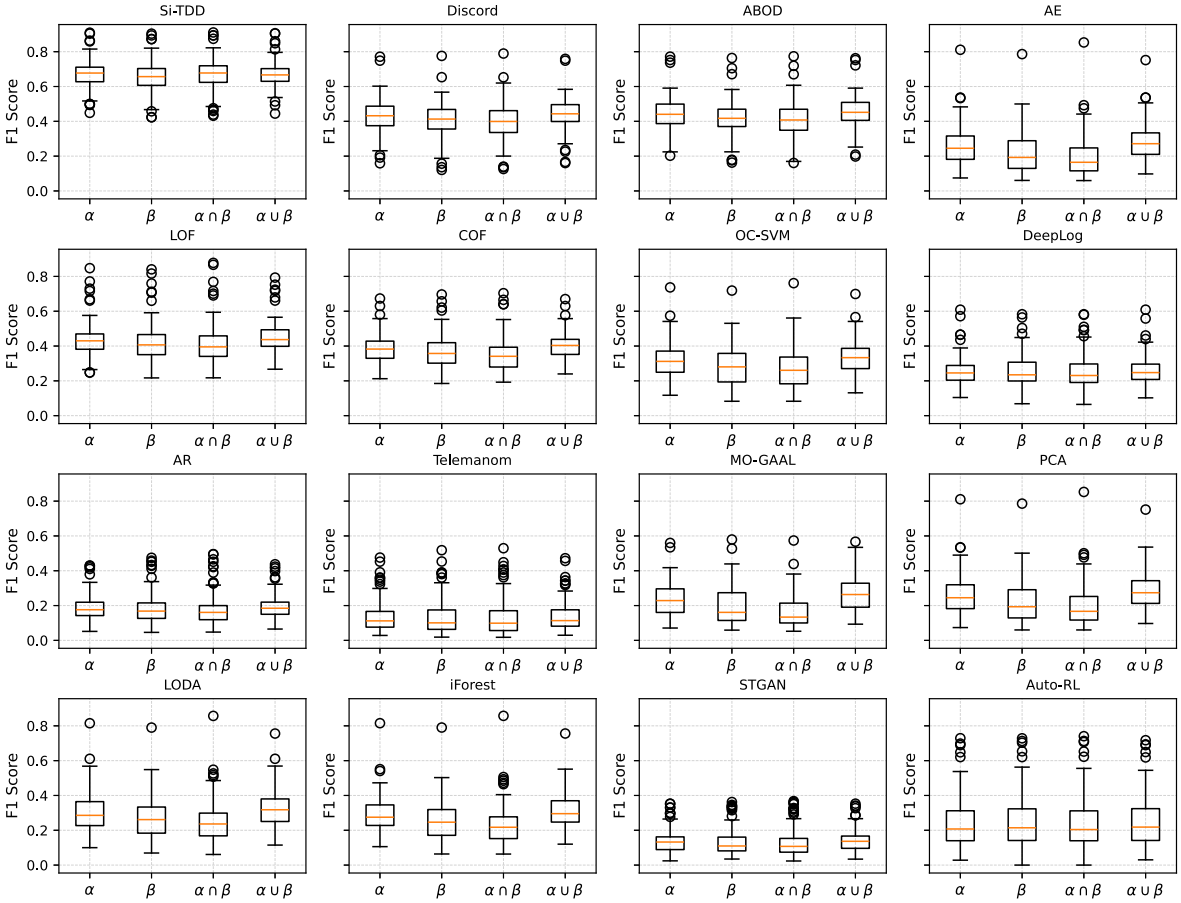


Fig. 12. Comparison of F1 score distributions across all locations for selected methods.

4.4. Si-TDD performance analysis

As outlined before, various types of anomalies are observable in traffic volume data by investigating deviations from the expected periodic patterns inherent in the data. Si-TDD leverages the same principle by quantifying the extent to which each data point deviates from other temporally similar observations, providing a robust mechanism for intensifying anomalies that disrupt these patterns. Fig. 15 visualizes the performance of Si-TDD across some cases from the AnoLT dataset. As shown, this method effectively identifies a wide range of anomalies in traffic data, from subtle point anomalies (Fig. 15 (b, d, f, and g)) to long and major deviations (Fig. 15 (c, e, and h)), requiring only one hyperparameter ($\mathcal{K}$) to be set. This contrasts with other state-of-the-art baselines, which often lack the capability to capture such nuanced anomalies and even major ones in some cases, while requiring various hyperparameters to be set. This requirement remains despite the fact that these methods, like Si-TDD, are ultimately expected to produce anomaly scores (Table 1) that still need to be converted into binary labels for practical use.

While Si-TDD demonstrates better alignment with the ground truth labels in AnoLT, it is shown that a perfect and complete match is not achieved yet. We investigated several situations where inconsistencies exist between the predicted anomalies and the labels in our data. Some examples of these cases, including certain False Positive (FP) and False Negative (FN) samples, are shown in Fig. 16. Based on this figure-and as observed in many other cases-all inconsistencies can be traced back to the *boundary point* concept mentioned earlier. Given our aim to meticulously label anomalies in volume data, this requirement was, in some cases, hard to fulfill. Due to the significant variation in data patterns between days of the week and across different locations, maintaining consistent labeling performance throughout the experiment proved challenging for the labelers. As a result, this led to some discrepancies both among the labelers and between their labels and the method outputs. Nevertheless, the intuitiveness and interpretability of Si-TDD provide valuable insights that enhance confidence in its performance in such cases. Si-TDD detects anomalies by identifying data points that are located sufficiently far from their respective nearest neighbors. The discriminative threshold is also constantly applied

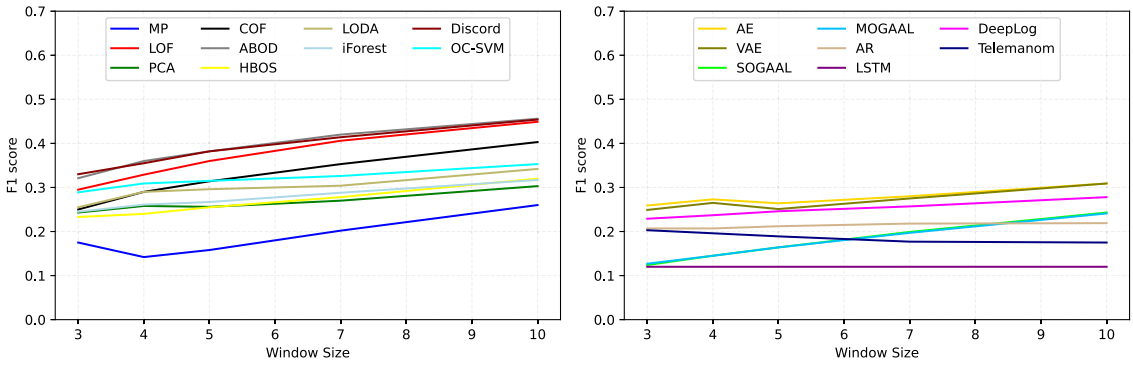


Fig. 13. F1 score variations by window-sizes in window-based TAD methods (based on α label set).

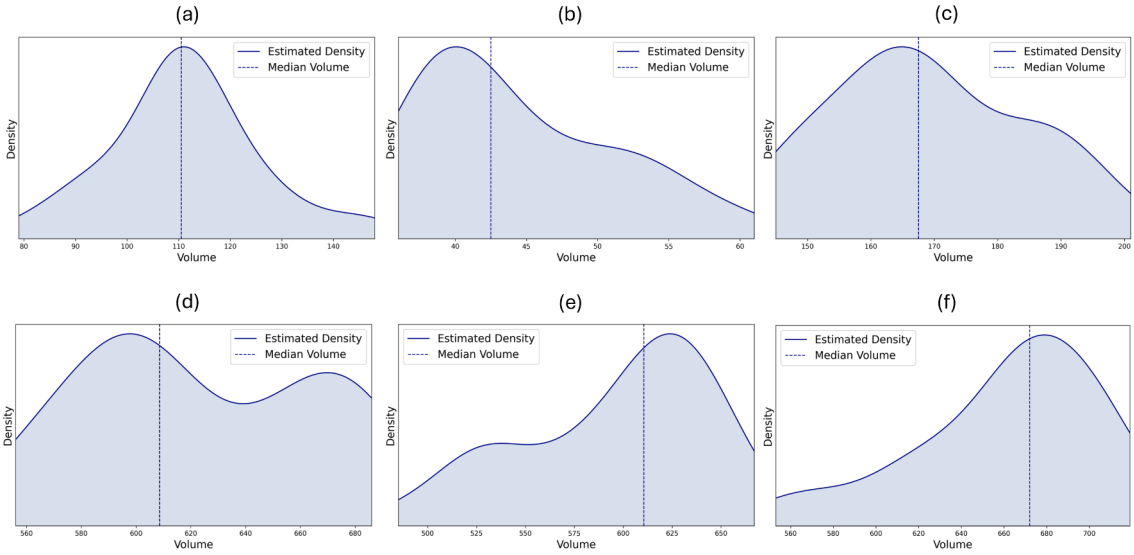


Fig. 14. Sample kernel density plots showing the distribution of traffic volume data across different time intervals on a specific day of the week at a specific location.

across each location. Therefore, any FP or FN sample can be attributed to inconsistencies in the labeler's standards during the labeling process, as also supported by the examples shown in Fig. 16.

It should be noted that anomalies labeled in the AnoLT dataset are intended to serve as representative labels for a completely fine-grained level of analysis. Si-TDD has demonstrated significant superiority over the baselines in generating anomaly scores that closely align with the labeled data. Anomaly labels at various levels of conservativeness can be produced by adjusting the thresholds applied in the last step. When applying thresholds for anomaly detection using Si-TDD, it is important to note the trade-off between the *early detection* of long subsequence anomalies and the identification of *extremely fine* point deviations. Prioritizing early detection can lead to the identification of more minor point anomalies that may have limited applicability in some downstream tasks. This is because any subtle anomalous point could potentially indicate the beginning of a long subsequence anomaly. However, such points may quickly revert to the normal range and merely represent a minor deviation with less practical significance.

Given Si-TDD's ability to assign continuous anomaly scores to data samples, the process of generating final anomaly labels for individual locations across a large network can also be made more efficient through guided automation. One practical approach involves visualizing a single representative data profile alongside its similar historical patterns for each location. Rather than requiring the user to label all anomalies, as described, the system can prompt the user to select or insert a single point they consider to represent the boundary between normal and anomalous behavior for a specific time interval. The score of this boundary point can then serve as a reference for that location, with all points scoring higher automatically labeled as anomalies. In addition, other techniques such as the knee-point method (Yu et al., 2019) can be utilized to determine natural inflection points in the sorted anomaly score distribution.

The sensitivity of the Si-TDD performance to its only hyperparameter, $\mathcal{K}$, as well as to different reference data sizes, is analyzed and shown in Figs. 17 and 18. $\mathcal{K}$ can be interpreted as the minimum number of observations required to establish a normal cluster within the data of each time interval. Firstly, the results in Fig. 17 indicate that the variation in performance with different values of

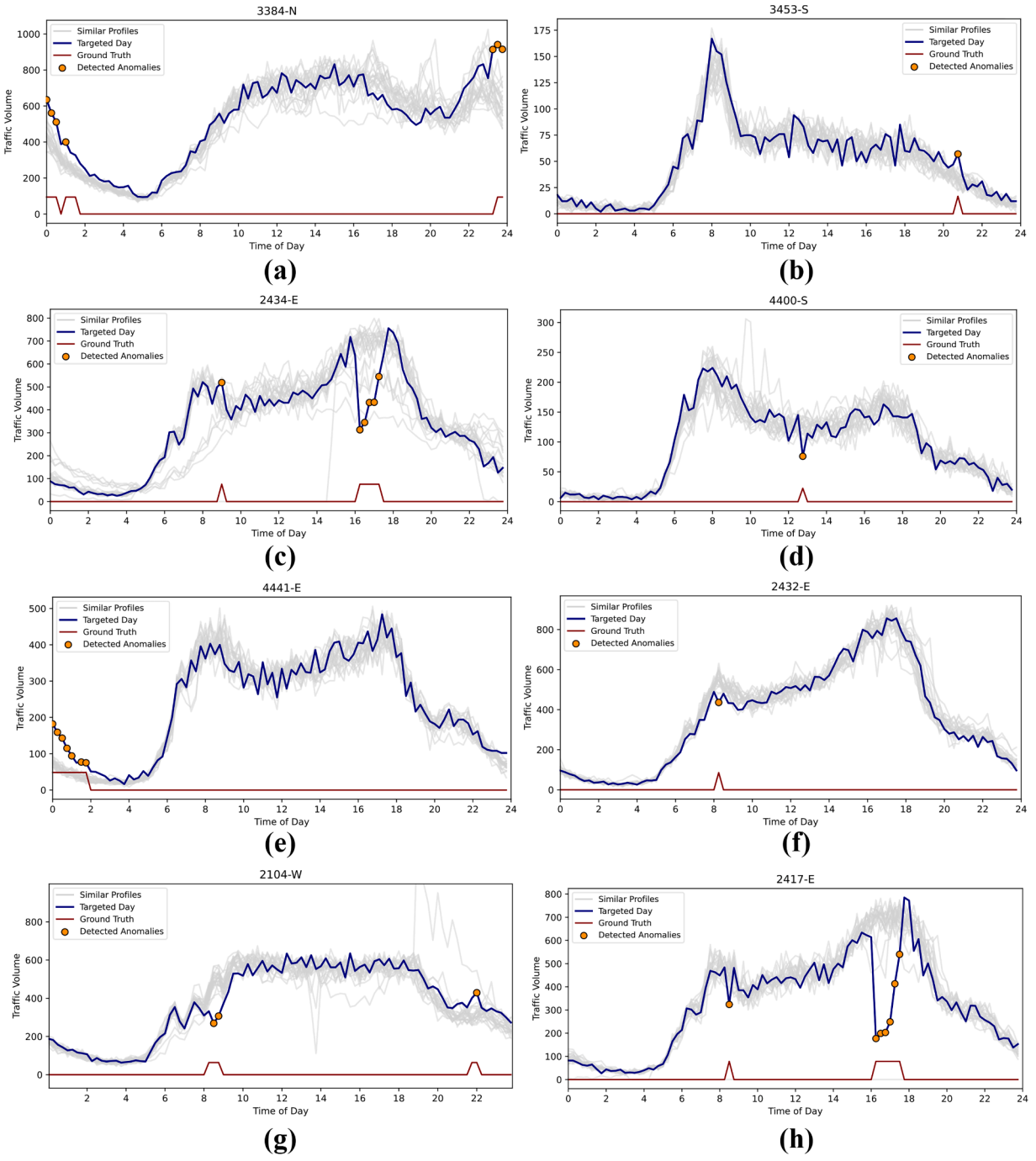


Fig. 15. Visualization of Si-TDD detection performance with the AnoLT dataset. The continuous red line at the bottom of each plot represents the α ground truth label set.

$\mathcal{K}$ is not substantial. Therefore, it can be concluded that Si-TDD is not *highly* sensitive to the value of this hyperparameter. However, the highest performance is recorded when $\mathcal{K} = 2$ or $\mathcal{K} = 3$ is used, with performance decreasing as the value of $\mathcal{K}$ moves further away from these values. Using $\mathcal{K} = 1$ is also not recommended in this context, as two similar anomalies may be found in close proximity and labeled as normal, an effect whose impact is experimentally recorded in Fig. 17. Furthermore, the trend in Fig. 18, obtained by decreasing the training data size, shows that Si-TDD maintains reasonable performance even with limited amounts of historical data. To be specific, using the same value of $\mathcal{K}$, a 60% reduction in reference data size (from five months to two months) decreased the F1 score from around 0.67 to 0.58, with almost similar behavior across different $\mathcal{K}$ values. However, this trend did not hold with one

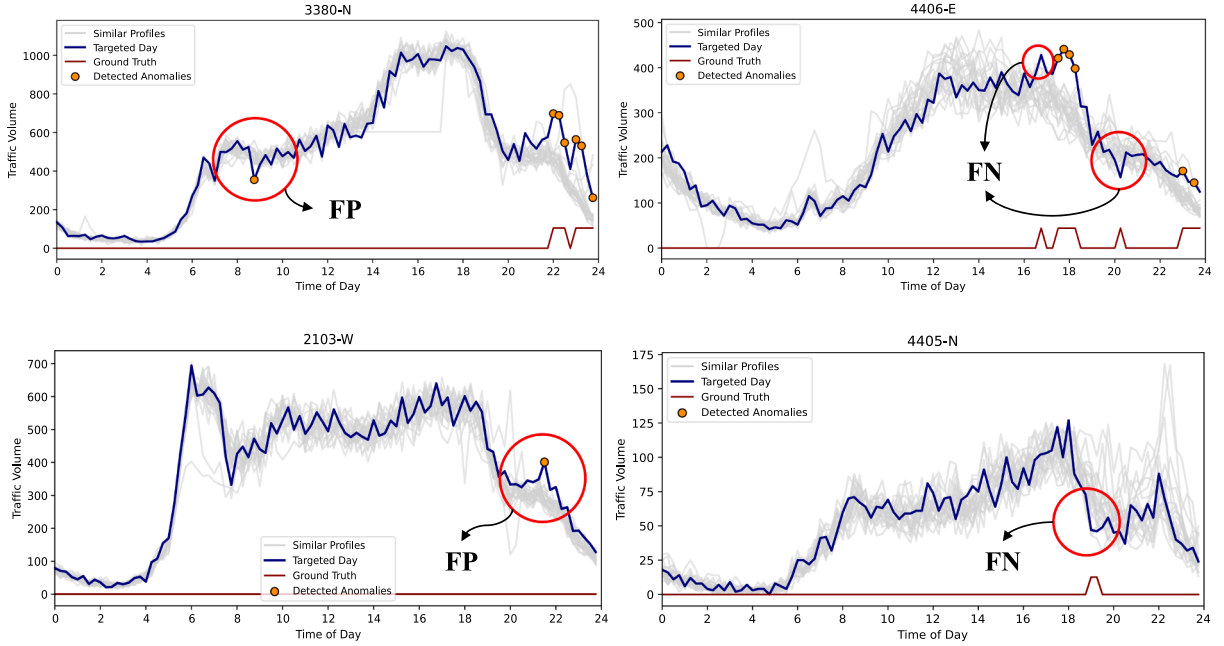


Fig. 16. Visualization of selected False Positives (FP) and False Negatives (FN) in the Si-TDD detection performance with the AnoLT dataset. The continuous red line represents the ground truth labels.

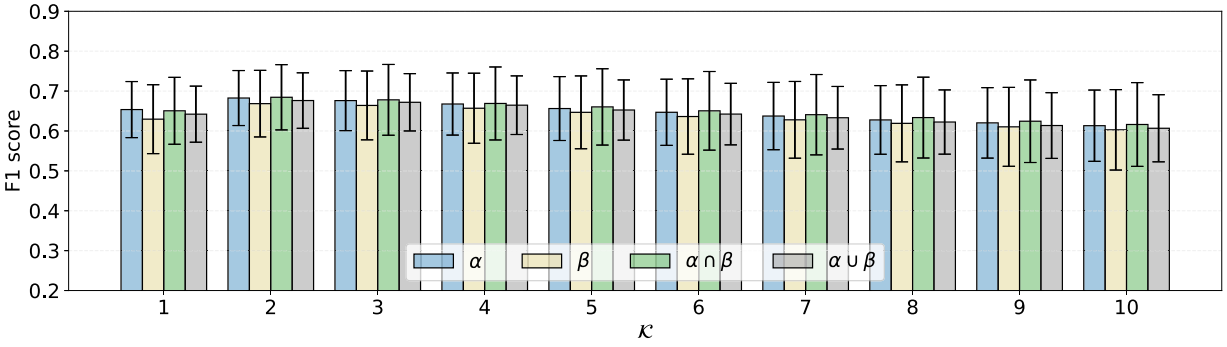


Fig. 17. Sensitivity analysis of performance with respect to $\mathcal{K}$ ($\mathcal{K}$th nearest neighbor to consider), the only hyperparameter in Si-TDD. Error bars indicate standard deviation.

month of reference data, showing a sharper drop and greater sensitivity to $\mathcal{K}$. The primary decreasing trend can largely be attributed to shifts in the *boundary* separating normal and abnormal data as the data size is reduced. As a number of labeled anomalies in AnoLT are subtle, a certain level of performance decline is expected when smaller reference datasets are used, and relabeling the ground truth anomalies based on the new reference may recover the loss in performance. Nonetheless, the results demonstrate that a large share of anomalies continue to be detected across different reference data sizes.

Based on the results in these figures, and considering the underlying concept of Si-TDD, using $\mathcal{K} = 3$ is recommended, provided that an adequate amount of historical data, such as three or more months, is available. Although $\mathcal{K} = 2$ delivered competitive performance in our experiments (even with one month of data), $\mathcal{K} = 3$ is a safer choice when more historical data are available. This is because two similar anomalies may occasionally appear in the training data at the same time step. If a new anomaly similar to these two also occurs, which is a rare situation, using $\mathcal{K} = 2$ may result in the method incorrectly classifying it as normal, whereas $\mathcal{K} = 3$ helps to avoid this issue. The impact of this issue for $\mathcal{K} = 1$ started to become evident with five months of data (Fig. 18), despite $\mathcal{K} = 1$ appearing competitive when four months of data or less are used. Therefore, a similar phenomenon may also occur for $\mathcal{K} = 2$ when larger amounts of historical data are considered or when the method is applied under new traffic conditions. With $\mathcal{K} = 3$, the presence of three neighboring samples, rather than two, is required to establish a normal region within the data. However, with limited data sizes of one month, $\mathcal{K} = 2$ is shown to provide better performance, but it should be noted that with such a data size, the full variation of the normal range is not adequately captured.

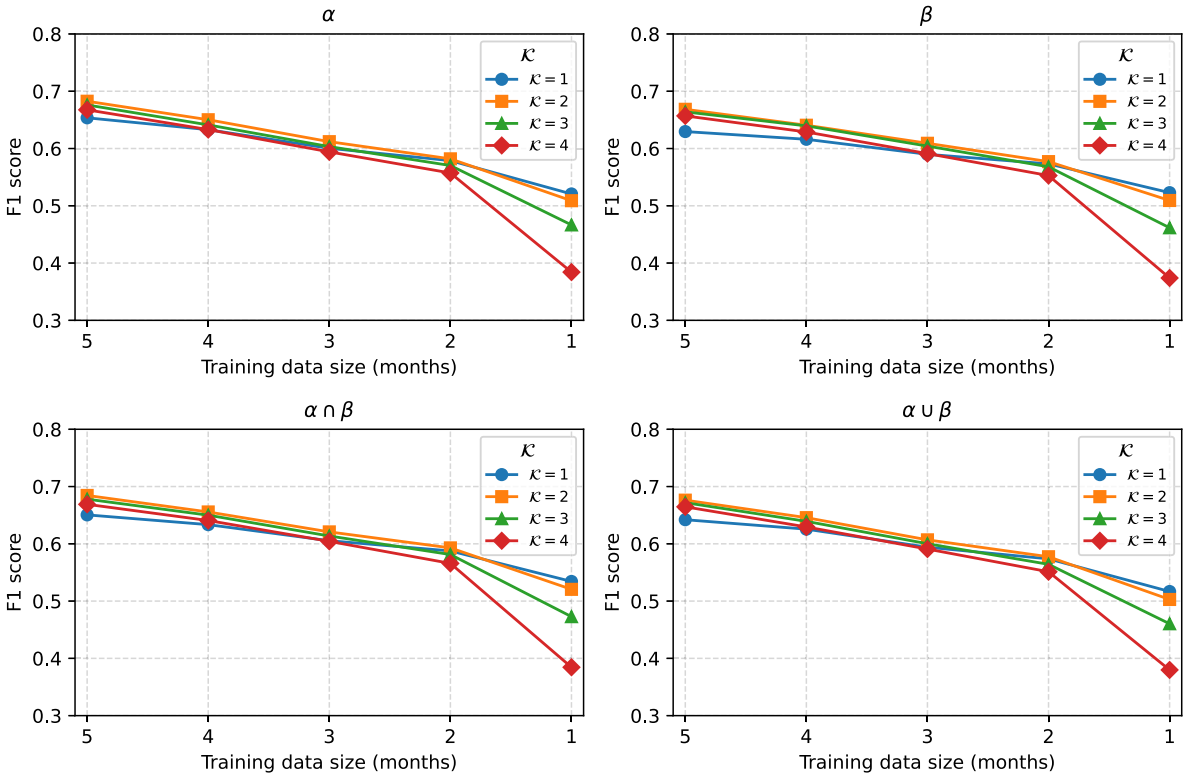


Fig. 18. Sensitivity analysis of Si-TDD performance with respect to reference (training) data size for varying $\mathcal{K}$ values under AnoLT label sets.

5. Summary and conclusion

In this paper, we investigate the applicability of state-of-the-art Time Series Anomaly Detection (TAD or TSAD) methods to real-world traffic volume data. First, the role of clearly defining anomalies and utilizing high-quality datasets in TAD is emphasized, highlighting how this has recently renewed interest in classical methods over newer alternatives. We then argue that these critical insights remain largely unexplored in the context of traffic and transportation data, with no clear conclusions drawn. To address this gap, we utilize a recent taxonomy of time series anomalies to elaborate on the anomaly types most likely to occur in traffic volume data and identify point and shapelet (subsequence) as the main categories requiring attention. Building on these insights, we design a traffic volume data monitoring platform to visually inspect and interactively label anomalies in real-world datasets. Inspired by time series discords-a state-of-the-art approach in TAD applications-we propose a customized, and lightweight method named Si-TDD (Step-isolated Traffic Discords Discovery), specifically tailored for detecting anomalies in traffic volume data. As part of this work, we also introduce AnoLT (Anomaly Labeled Traffic), an open-source dataset comprising anomaly-labeled road volume data collected from 147 locations across two urban networks in Melbourne, Australia. Comparative studies against more than 20 state-of-the-art TAD methods with the AnoLT dataset show that Si-TDD significantly outperforms existing solutions, achieving an F1 score of 67% for fine-grained traffic volume anomaly detection.

Based on the discussion and results, the following items can be listed as the key findings of this work:

1. Existing anomaly-labeled road traffic datasets often rely on reported incidents or public records. We argued that anomalies go beyond these events and relying solely on such information as ground truth labels may not capture the full scope of anomalies. Therefore, alternative perspectives on ground truth labeling need to be explored.
2. Developing effective anomaly detection solutions requires a clear understanding of the targeted anomalies and their unique characteristics. This clarity is crucial for achieving successful outcomes in TAD applications, as demonstrated in this paper.
3. AnoLT dataset provides a more comprehensive and nuanced foundation for evaluating detection solutions using traffic volume data. It combines fine-grained anomaly labels with real-world volume observations, enhancing its reliability and usability for this domain.
4. Although classical TAD methods exhibit limited effectiveness in their original form when applied to traffic data, this study demonstrates that, with contextual adaptation, they can be refined into highly competitive and reliable solutions.

5. The proposed Si-TDD method (step-wise implementation of discord discovery in volume time series data) is established as a state-of-the-art solution for traffic volume anomaly detection. Si-TDD showed robust and intuitive detection capabilities, significantly outperforming existing TAD baselines from the literature.
6. Consistent with recent benchmarking studies, classical methods remain viable and can achieve state-of-the-art performance in traffic volume anomaly detection. For future solutions, it is highly recommended to explicitly discuss the advantages offered over such well-established, intuitive, and straightforward approaches.

Aligned with previous works and empirical evidence, this study is grounded in the perspective that traffic data anomalies are those segments of data that considerably deviate beyond the typical variation observed in available historical traffic patterns. Future work could investigate whether certain anomalies are embedded within normal traffic variation, making them difficult to detect visually or distinguish algorithmically under the current framework. Furthermore, it is also valuable to explore the extent to which underlying changes in data patterns over time (e.g., concept drift) can affect anomaly detection performance in practice, and whether regular updates to the training data can properly address this issue. If so, identifying suitable parameter settings such as update frequency or base data size would become another relevant area of investigation. Another potential avenue of research can be built upon the findings of this study by exploring solutions for one-time threshold calibration across an entire network. Such an approach would aim to address the variability in scale observed across different locations while reducing the need for location-specific tuning. Although existing literature often relies on external sources such as incident reports or prior knowledge to generate binary labels, the findings of this paper suggest that an effective threshold search space can be derived directly from each individual time series. Further investigation could explore whether dynamic thresholding strategies can generalize the strictness level calibrated for one location to others across the whole network.

CRediT authorship contribution statement

Iman Taheri Sarteshnizi: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Majid Sarvi:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Conceptualization; **Saeed Asadi Bagloee:** Writing – review & editing, Supervision, Software, Methodology, Investigation, Conceptualization; **Neema Nassir:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

Data availability

A link to the data is provided.

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